

Neural Representations of Perceived Engagement during Action Observation

Edoardo Arcuri (*)¹, Leonardo Cerliani², Martina Ardizzi¹, Nunzio Langiulli¹, Francesca Ferroni¹, Christian Keysers^{2,3,+}, Valeria Gazzola^{2,3,+}, Vittorio Gallese^{1,4,+}.

1. Department of Medicine and Surgery, Unit of Neuroscience, University of Parma, Parma, Italy

2. Netherlands Institute for Neuroscience, Royal Netherlands Academy of Art and Sciences, Amsterdam, Netherlands

3. Department of Psychology, University of Amsterdam, Amsterdam, the Netherlands

4. Italian Academy for Advanced Studies in America at Columbia University, New York, USA

+: These authors contributed equally

Corresponding author (*):

Edoardo Arcuri; Via Volturmo, 39/E, 43121, Parma (Italy). Telephone: +39-0521-903873; E-mail: edoardo.arcuri@unipr.it

Abstract

Interpersonal motor interactions are central to social life, yet it remains unclear how social cues relevant to detecting engagement are encoded in the brain. Recent evidence suggests that regions traditionally associated with mentalizing, such as the dorsomedial prefrontal cortex (dmPFC) and temporo-parietal junction (TPJ), co-activate with nodes of the Action Observation Network (AON) during motor engagement with others, pointing to a synergistic role in the processing of action features during interaction. Using fMRI and Representational Similarity Analysis (RSA), we examined brain responses to reach-to-grasp actions varying in Goal (passing vs. placing), Perspective (2nd vs. 3rd person), and Gaze visibility, creating a gradient of perceived engagement. Our results show that the TPJ and premotor cortices, but not the dmPFC, displayed convergent neural geometry during action observation. Model- and cluster-based analyses further linked TPJ and AON regions in multiple representational spaces: left-hemisphere regions aligned with goal encoding, whereas right-hemisphere regions aligned with an Engagement model capturing first-person interaction. Critically, we demonstrate that the rTPJ and left premotor cortex share representational geometry for action perspective, directly coupling mentalizing and sensorimotor systems in the encoding of directional cues. By contrast, the dmPFC appeared representationally isolated from both TPJ and AON regions, with response patterns showing no similarity to Goal or Engagement models. This is consistent with accounts of its recruitment under richer interactive demands rather than selectivity for abstract action features. Together, these findings support a distributed, sensorimotor account of engagement encoding and reveal new functional links among key social cognition areas.

Keywords: Social Cognition; Action Observation; Engagement; RSA; Mirror Neuron System; TPJ.

1. Introduction

When in the presence of other agents, communicative or motor actions directed toward the observer form a special class of social events, as they imply potential direct involvement in interaction. Such engagement can be inferred from bodily cues such as movements, gaze, and communicative gestures, which often occur on very fast timescales and require prompt interpretation and response. Despite decades of research on the neural substrates of social cognition, however, how the brain encodes potential engagement with others remains largely unclear.

In the human brain, two systems have traditionally drawn interest for processing engagement-relevant features. The first is the Action Observation Network (AON), centred around parieto-frontal circuits, active during observation and, in some nodes, execution of motor actions (Molenberghs et al., 2012; Rizzolatti & Sinigaglia, 2016). The second is the so-called ‘Mentalizing’ Network (MENTA), including regions such as the dorsomedial prefrontal cortex (dmPFC) and temporo-parietal junction (TPJ), consistently associated with Theory of Mind tasks (Molenberghs et al., 2016; Saxe & Wexler, 2005; Schurz et al., 2014).

These networks are often described as supporting qualitatively different types of social information (Alcalá-López et al., 2019; Geiger et al., 2019; Spunt et al., 2011; Wheatley et al., 2007). The AON is typically linked to “low-level” sensorimotor features such as kinematics and motor goals (Cattaneo et al., 2009; Giese & Rizzolatti, 2015), whereas the MENTA is associated with “high-level” inferences about intentions and affective states (Frith & Frith, 2021; Saxe, 2006; Thornton et al., 2019).

Yet, this dichotomy is increasingly challenged, as evidence shows that both systems are engaged across multiple levels of social cognition (Gallese, 2007; Schmidt et al., 2021; Schurz et al., 2020). For example, AON parieto-frontal regions are recruited when intentions and emotions are inferred from bodily kinematics (Boccardo et al., 2019; Patri et al., 2020; Spunt et al., 2011) and host ‘mirror mechanisms’ implicated in joint action and affective alignment (Bonini et al., 2022; Del Vecchio et al., 2024; Ferrari-Toniolo et al., 2019). Conversely, the MENTA often co-activates with premotor regions during motor interactions, even in the absence of explicit mentalizing demands (Newman-Norlund et al., 2008; Schilbach et al., 2010).

Moreover, “mentalizing” areas such as the dmPFC and TPJ participate in a wide variety of processes beyond social cognition. The dmPFC contributes to semantic memory and executive control (Kolling & O’Reilly, 2018; Lieberman et al., 2019; Narayanan & Laubach, 2006), while the TPJ is implicated in multisensory integration and visuospatial processing (Blanke & Arzy, 2005; Corbetta et al., 2008; Ionta et al., 2011). This functional heterogeneity complicates the assumption of a purely “high-level” role in social reasoning and leaves their precise contribution to social cognition unresolved.

An alternative view proposes that the AON and MENTA are complementary, sharing multimodal, agent-related representations (Keysers & Gazzola, 2007; Kokal et al., 2009; Miss et al., 2022; Ninomiya et al., 2018). Studies in interactive settings show that monitoring others’ actions or gaze during coordination increases functional connectivity between motor regions and both the dmPFC (Ciaramidaro et al., 2014; Wang et al., 2011) and TPJ (Miyata et al., 2021; Sperduti et al., 2014), suggesting that these areas operate synergistically during sensorimotor control in social interaction. Consistent with this, both networks are modulated by interactive contexts and by observed actions affording direct engagement. For instance, frontoparietal AON regions and the mesial prefrontal cortex respond more strongly to socially directed actions than to non-social ones (Becchio et al., 2012; Maffei et al., 2020; Möttönen et al., 2016), with the dmPFC showing heightened sensitivity to actions and expressions directed at the observer (Nagels et al., 2015; Shepherd & Freiwald, 2018; Sliwa & Freiwald, 2017).

Specific features accompanying these social signals are known to modulate neural activity in both AON and MENTA regions. For example, sensorimotor areas encode the same action differently when it is directed toward the observer (Angelini et al., 2021; Bonini et al., 2014), and recent findings show that kinematic cues relevant for coordination with others are more salient to observers when viewed from a second-person

103 perspective (Arcuri et al., 2025). Alongside motor features, gaze cues are especially relevant to determine
104 engagement, as demonstrated by consistent recruitment of the dmPFC in gaze and attention monitoring in
105 face-to-face contexts (Bristow et al., 2007; Dal Monte et al., 2022; Hayashi et al., 2020) and by its stronger
106 activation by simple eye contact (Cavallo et al., 2015; Kampe et al., 2003). Superior temporal regions, sensory
107 relays of the AON, are likewise sensitive to gaze direction, showing enhanced activity for direct eye contact
108 (Jellema et al., 2000; Marquardt et al., 2017; McGettigan et al., 2017). Studies measuring motor-evoked
109 potentials (MEPs) reveal that sub-threshold motor activity increases during action observation when eye
110 contact is maintained (Bekkali et al., 2022; Prinsen & Alaerts, 2019), indicating that gaze cues modulate
111 premotor processing linking them to action preparation (see also (Coudé et al., 2016; Maranesi et al., 2013)).
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113 The TPJ, another key MENTA node, has been implicated in the automatic processing of self- and other-related
114 visual perspectives (Guterstam et al., 2021; Martin et al., 2020), likely supporting perspective-taking during
115 interaction (Igelström & Graziano, 2017; Krall et al., 2015). Supporting this, stimulation studies demonstrate
116 that TPJ activity can modulate both self-other perspective distinction and preparatory motor responses to
117 observed actions (Bardi et al., 2017; Hogeveen et al., 2015; Reader et al., 2018). Moreover, neural activity in
118 both premotor regions and the TPJ scales with the degree of social coordination required during motor tasks,
119 when efficient monitoring of self and other actions is essential (Abe et al., 2019; Cheng et al., 2022; Saito et
120 al., 2010; Tognoli et al., 2007).
121

122 Altogether, this evidence suggests that both the AON and MENTA contribute to monitoring and encoding
123 potential engagement with others, pointing to a shared role in interactive contexts and possibly overlapping
124 representational formats. In the present study, we tested this hypothesis using fMRI and Representational
125 Similarity Analysis (RSA). Participants passively observed grasping actions varying in engagement level while
126 we recorded fMRI data. RSA, based on multivoxel pattern analysis (Haxby et al., 2001) allowed us to
127 characterise the structure of neural activity patterns within regions and examine the similarity of
128 representational geometries across them.
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130 We pursued three complementary aims. First, we assessed ‘representational connectivity’ (Kriegeskorte,
131 2008) by performing a neural RSA, testing whether engagement-related modulation of BOLD patterns was
132 shared across regions. This analysis captures how distributed regions coordinate their representational
133 formats during social perception.
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135 Second, we asked whether established building blocks of second-person contexts could explain the observed
136 representational geometries. Prior studies have typically manipulated one of three features: the goal of the
137 observed action, its orientation (toward or away from the observer), and its social signalling through gaze or
138 eye contact. While each has been shown to modulate neural responses individually, their combined influence
139 on representational geometry and inter-regional coupling remains poorly understood. To address this, we
140 performed a model-based RSA, comparing neural dissimilarity matrices to idealised models for these
141 features.
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143 Finally, we tested whether engagement itself is represented in a distinct format. We constructed a fourth
144 model based on participants’ explicit ratings of perceived engagement, allowing us to ask whether
145 engagement constitutes a discrete representational feature in its own right, or whether it instead arises from
146 the integration of bodily cues across AON and MENTA.
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2. Materials and Methods

The study was approved by the local ethics committee (Comitato Etico dell'Area Vasta Emilia Nord, protocol n. ID SIRER 5476 - 905/2022) and was conducted following the principles expressed in the Declaration of Helsinki. All participants provided written informed consent.

2.2. Participants

26 naïve volunteers (13 males and 13 females, mean age 24.3 ± 3.4 years; range 18-36) were recruited via the University of Parma mailing channels, social media, and flyers distributed across the University campuses. All the tasks in the study were administered in Italian, except for an anglophone participant whose given instructions (both verbal and written) were in English. All participants were right-handed according to the Edinburgh Handedness Inventory, had normal or corrected-to-normal vision, and had no previous diagnosis of neurological or psychiatric disorders as assessed by self-report.

2.3. Stimuli

The fMRI task was an observation paradigm using video clips recorded in a previous study (Arcuri et al., 2025). Participants viewed 2-second clips of different individuals, hereafter referred to as 'agents', seated at a table, grasping a wooden cube either to pass it to someone (PASS) or to place it on a pedestal (PLACE).

Both actions began with the agent's arm resting on the table, hand in a pinch position, and gaze directed at the object. PASS and PLACE actions also shared an initial reach-to-grasp phase, during which the cube was grasped while gaze remained on it. In PLACE actions, the object was lifted, placed on the pedestal, and released; videos ended 500 ms after release. In PASS actions, the object was lifted, the arm fully extended toward a hidden recipient, and eye contact established; videos ended 500 ms after full extension. Each action was filmed from two angles: front-facing and 45° left, the latter obscuring the receiver.

Clips were shown from either a third-person (agent facing away) or second-person (facing the observer) perspective. To manipulate gaze, actors wore transparent eyeglasses, digitally edited in a second version to appear as opaque sunglasses. This yielded a $2 \times 2 \times 2$ factorial design with factors: GOAL (PASS vs. PLACE), PERSPECTIVE (2nd vs. 3rd person), and GAZE (visible vs. obscured). Each factor contrasted an engaging condition (PASS, 2nd-person, gaze visible) with a non-engaging one (PLACE, 3rd-person, gaze obscured). In this way, the full design created a gradient of engagement, from least engaging ("PLACE-3rd-no gaze") to most engaging ("PASS-2nd-gaze"), with intermediate combinations capturing the relative weight of each feature (Figure 1A).

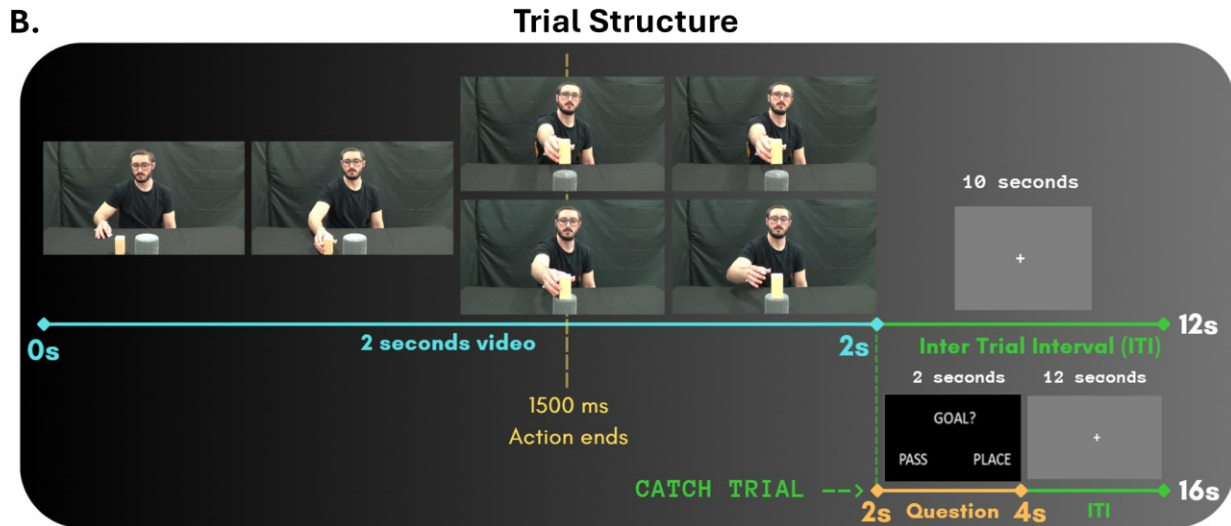
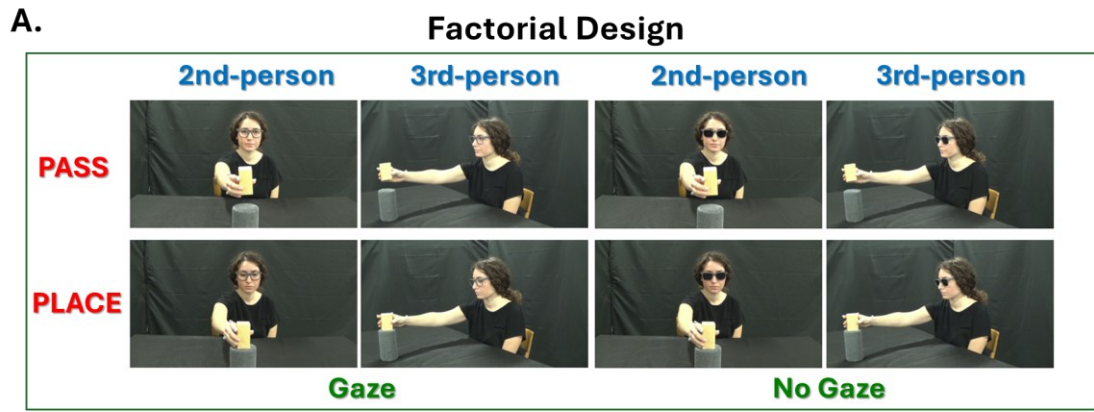


Figure 1 | Video stimuli used in the study. **(A)** Factorial design (2 x 2 x 2) of used conditions. **(B)** Structure of single observation and catch trials used in the study.

8 actions were selected among the 96 available recorded actions from (Arcuri et al. 2024), resulting in 1 grasping action for placing and 1 grasping action for passing from 4 different actors (total: 4 unique passing actions, 4 unique placing actions), balanced for sex (i.e., 2 males and 2 females). For each unique action, both first and third-person perspective recordings were used, and each video clip underwent the editing procedure for gaze covering. In total, 32 action stimuli were created, 4 for each condition of the factorial design. To select the 8 action stimuli among the 96 available, we verified which among the prototypical PASS and PLACE actions employed in the action prediction studies were predicted with the highest accuracy (i.e., % of correct responses), mediating by scores obtained by the action prediction studies employed in (Arcuri et al. 2024). Following this procedure, the 2 actors among the 4 female actors available and 2 actors among the 4 male actors available showing the highest scores were selected.

Finally, a ninth NON-SOCIAL condition was added, consisting of 2 clips showing only the object (no agent) from the two perspectives. In total, 34 video stimuli were used (32 action, 2 non-social).

2.4. Experimental design and procedure

A) fMRI session.

The experimental session consisted of four fMRI runs, each lasting about 6.5 minutes, for a total of 30–32 minutes including breaks. Stimuli were presented in an event-related design, with a fully randomized order within each run. Videos were displayed through MRI-compatible binocular Full HD OLED goggles (Nordic

NeuroLab, Norway), and responses were collected using an MRI-compatible Joypad, programmed in MATLAB R2022b.

Each run included 38 trials: 34 observation trials and 4 catch trials (10%). The observation trials featured one of the 32 unique action videos, each presented once per run. After each video, a black screen with a fixation cross was shown for a fixed 10 s inter-trial interval (ITI), yielding a stimulus-onset asynchrony (SOA) of 12 s (Figure 2). Note that the jittering procedure on ITI, usually indicated for optimizing BOLD detection in event-related design (cf. Buckner 1998; Petersen and Dubis 2012), was not adopted here, as previous studies systematically investigating the effect of jittered ITI on stimulus modelling for MVPA analysis, involving modelling at the single stimulus level, showed no significant improvement in analysis results by employing jittered designs (cf. Zeithamova et al., 2017; Popal et al. 2019). Since the action videos were shown only once per run, this resulted in 128 trials to be used for planned analyses, with 4 repetitions per condition for first-level modelling in both multivariate and univariate analyses (see Section 2.7). Catch trials (16 in total) featured randomly selected action videos, followed by a 2 s response window in which participants were required to indicate the goal (pass or place) of the observed action. These trials used a 12 s ITI, for an SOA of 16 s (Figure 1B).

Before scanning, participants were instructed to attend carefully to each action video, maintain fixation when the cross was visible, and respond during catch trials by pressing with either the thumb or index finger on the joystick. This ensured attention monitoring and maintained engagement throughout the task.

B) Post-fMRI survey: Engagement Evaluation Questionnaire (EEQ)

Following the fMRI session, participants completed an online questionnaire created with PsychoPy (Peirce et al., 2022) and administered via Pavlovia.org. They were asked to re-evaluate the 32 action stimuli from the experiment, rating specific parameters on a continuous scale (1 to 100; values were hidden to participants). In the questionnaire, each video was presented once, followed by a request to provide a rating ranging from “Not at all” (Italian version: “per niente”) to “A lot” (Italian version: “molto”) for the following questions:

- Question 1 (Q1): «How communicative does the action seem to you?» (Italian: “Quanto ti sembra comunicativa l’azione?”)
- Question 2 (Q2): «How potentially involved do you feel in the action observed?» (Italian: “Quanto ti senti potenzialmente coinvolto nell’azione osservata?”)
- Question 3 (Q3): «How much does the video make you feel observed?» (Italian: “Quanto ti fa sentire osservato il video?”)

The responses to the three EEQ questions were used as a measure of explicit engagement perception, with each question mirroring the three levels of our factorial design (i.e., Q1: Goal, Q2: Perspective, Q3: Gaze).

To evaluate the consistency of participants’ ratings across Q1–Q3, we used a multivariate distance approach. For each video, we treated each participant’s responses as a three-dimensional vector [Q1,Q2,Q3] and computed its Mahalanobis distance to the group centroid (Mahalanobis, 1936). Distances were evaluated against the χ^2 distribution with 3 degrees of freedom (critical value $\chi^2(0.975,3) = 11.34$ for squared distances, corresponding to ≈ 3.37 for the distance itself), with participants exceeding this cutoff flagged as outliers (see (De Maesschalck et al., 2000), for an overview). This multivariate approach was preferred over univariate measures as it accounts for the covariance structure among Q1, Q2, and Q3, weighting deviations according to how the three scores vary together. All computations were carried out in R (2021) using the built-in mahalanobis() function.

After this evaluation, ratings from the questionnaire were subsequently used to build the ‘Engagement Rating’ dissimilarity matrices for comparison with neural patterns (see Section 2.7).

2.5. Image acquisition

Structural T1-weighted and functional T2*-weighted images were acquired on a 3T General Electric SIGNA Premier wide-bore scanner with a 48-channel head coil. Functional BOLD images were obtained using a gradient-echo echo-planar imaging (EPI) sequence with multiband factor 2. Acquisition parameters were: 70 axial slices (whole-brain coverage), interleaved bottom-up order, 2.0 mm isotropic voxels, slice thickness 2.0 mm (no gap), TR = 2000 ms, TE = 30 ms, FOV = 256 mm, flip angle = 90°. Due to a recording error, phase-encoding direction varied across participants: 15 were acquired with antero-posterior (AP) encoding and 11 with postero-anterior (PA) encoding. Field distortion correction was therefore included in preprocessing (see Section 2.6).

A high-resolution 3D T1-weighted MP-RAGE volume was acquired for anatomical reference, covering the whole brain. Parameters: 256 axial slices, 1 mm isotropic resolution, FOV = 256 mm, TI = 1100 ms, TE = 3.1 ms, TR = 2070 ms, flip angle = 8°.

2.6. Image preprocessing

Functional image preprocessing was performed using FEAT (FMRI Expert Analysis Tool, cf (Jenkinson et al., 2012)) v6.00, implemented in FSL (<https://fsl.fmrib.ox.ac.uk/fsl>), version 6.0.7. For structural processing, the software ANTs (Advanced Normalization Tools, <https://stnava.github.io/ANTs>) was used for brain extraction and intensity nonuniformity correction, while FSL's package FAST (FMRIB's Automated Segmentation Tool) was used for tissue segmentation.

All anatomical T1-weighted images underwent the following preprocessing steps: correction for radiofrequency (RF) intensity and skull-strip employing the software ANTs (command 'antsBrainExtraction.sh' implemented on personal bash scripts); segmentation of brain tissues (cerebrospinal fluid CSF, White matter WM, and Grey matter GM) on the corrected and brain-extracted T1-weighted images using FSL's package FAST; visual check for quality evaluation.

Before proceeding with the full preprocessing pipeline for functional images, a synthetic, undistorted BOLD image for each run of each participant was generated with the tool SynBOLD-DisCo (Synthetic BOLD contrast for Distortion Correction, Yu et al. 2023). Synthetic images were produced using the preprocessed T1-weighted image and the raw 4D BOLD images as inputs. The synthetic, undistorted BOLD images were used together with the original distorted BOLD images as input to the FSL package TOPUP to perform correction of spatial distortions due to magnetic field susceptibility on the functional images. This step was regarded as necessary for two main reasons: first, obtaining an overall better quality of voxel displacement in the brain's prefrontal areas, in which one of the ROI is located (dmPFC); second, and most importantly, to uniform our sample, which suffered from having two different PE directions.

After distortion correction, functional images underwent the following preprocessing steps: motion estimation and correction through rigid body transformation with 6 degrees of freedom (DOF), MCFLIRT package (Jenkinson et al., 2002); slice timing correction using Fourier-space time-series phase-shifting; co-registration (6 DOF) of BOLD images to the standardised T1-weighted (12 DOF, FNIRT package, MNI128 template as reference) with FLIRT boundary-based registration (BBR); spatial smoothing using a 4mm FWHM Gaussian kernel; grand-mean intensity normalisation of the entire 4D dataset by a single multiplicative factor; High pass temporal filtering (Gaussian-weighted least-squares straight line fitting, with sigma=30.0s) for removing low frequency.

2.7. fMRI data analysis

All statistical analyses on fMRI data were carried out using FSL's FEAT. R (2021) was used for analysing ratings and performing RSA on both behavioural and fMRI data.

A) General Linear Model (GLM)

Voxel-wise response amplitudes to experimental conditions of interests were estimated with a GLM, and beta maps were converted to z-scores. The model included 11 predictors: 8 regressors for action videos ($2 \times 2 \times 2$ factorial design); 2 regressors for catch trials (video and response window); 1 regressor for non-social videos. All events were modelled as 2 s boxcars from onset, convolved with a canonical HRF (Gamma function; (Lindquist et al., 2009)). Temporal derivatives were added to account for hemodynamic timing variability, particularly relevant for short TR designs (Sladky et al., 2011). Six head-motion parameters (3 translations, 3 rotations) were included as nuisance regressors. Temporal autocorrelation was addressed using FSL's pre-whitening (M. W. Woolrich et al., 2001)-

Each regressor was first contrasted against baseline to obtain its voxel-level activation profile. The resulting beta maps for the eight action conditions served as input for the RSA (see next section). Although RSA was the primary analytical focus of this study, we additionally ran planned whole-brain univariate contrasts to assess differences in BOLD activation between engagement-relevant (PASS, 2nd-person, visible gaze) and non-engaging features. For each factor (GOAL, PERSPECTIVE, GAZE), linear contrasts were computed in both directions (e.g., PASS vs. PLACE and PLACE vs. PASS) by aggregating across the relevant conditions (e.g., PASS vs. PLACE assigned positive weights to all PASS conditions and negative weights to all PLACE conditions, yielding a $[1 \ 1 \ 1 \ 1 \ -1 \ -1 \ -1 \ -1 \ 0 \ 0 \ 0]$ contrast vector). Within-subject estimates were obtained using FSL's fixed-effects model (FLAME, random-effects variance forced to zero), and group-level results were analysed with FLAME 1+2 mixed-effects, accounting for both within- and between-subject variability (M. Woolrich, 2008). Cluster-based inference was applied with a cluster-defining threshold of $z = 3.1$ and a corrected cluster significance of $p < 0.05$ (FWER). It is important to note that these univariate analyses provide only a limited view of the data, given that the experimental design was optimized for multivariate RSA rather than for eliciting peak activations. In particular, the relatively small number of repetitions and the event-related design reduce sensitivity for detecting peak activations, compared to block designs and broader stimulus sets used in prior work (e.g., Becchio et al., 2012; Ciaramidaro et al., 2014). Full results are reported in the Supplementary Materials (page 2).

Lastly, to rule out potential effects of PE direction, subjects' z-maps of each factorial condition against baseline were compared using FSL's *randomise* (5000 permutations, two-sample t-test). No significant group differences were observed for any contrast at the conventional alpha level. For completeness, we report in the Supplementary Materials (page 6) the lowest t- and p-values at which at least one cluster reached significance.

B) ROIs' RSA

To examine the representation of engagement-related features in MNS and MENTA ROIs, we performed a Representational Similarity Analysis (RSA). RSA represents the activity profile of an ROI in terms of the similarity of its neural patterns across experimental conditions, referred to as first-level representational similarity (cf. Kriegeskorte et al., 2008). This is achieved by pairwise comparisons of neural patterns between conditions, yielding a representational dissimilarity matrix (RDM) that encodes distances according to a chosen metric. The resulting neural RDM describes the representational structure of a ROI and can be compared with other RDMs for second-level similarity analysis. Other RDMs may be derived from neural activity in different ROIs, behavioural ratings, or conceptual models of hypothesized representational structures (Popal et al., 2019).

First-level RSA (Representational Dissimilarity Matrices, RDMs).

Neural RDMs. For each subject, we computed ROI-specific RDMs from normalised beta maps obtained in the factorial GLM, which produced one regressor per condition. Vectors of beta values were extracted from ROI voxels for each condition, and $1-r$ (Pearson correlation) was used as the distance metric, producing an 8×8 RDM across the eight action conditions.

Conceptual RDMs. To test the representational structure of neural patterns, we constructed three conceptual RDMs based on the factorial design: GOAL, PERSPECTIVE, and GAZE. In each, the engagement-relevant feature was coded as maximally dissimilar (1) from its non-engaging counterpart (0). Resulting RDMs are shown in Figure 2A (red, blue, and green matrices).

Engagement Ratings RDM. We also computed an RDM from participants' EEQ scores to capture perceived engagement. As described in section 2.4, for each subject, Mahalanobis distances were calculated between condition-wise vectors of average ratings across the three questions. Mahalanobis analysis revealed that participants' ratings were highly consistent overall, as the mean distances per video were low ($M = 1.53$, range = 1.49–1.57), well below the χ^2 cutoff (≈ 3.37) (for full results, see the supplementary data 1). Outliers were rare, averaging fewer than one participant per video ($M = 0.88$, range = 0–1), and were not systematic across stimuli, suggesting that variability reflected stimulus-specific ambiguity rather than unreliable raters.

The Engagement Ratings RDM was then computed by calculating the Mahalanobis distance to assess between-condition dissimilarities. At the group level, a clear categorical separation between PASS–2nd-person conditions (with or without gaze) and all other conditions emerged (see figure 2A, pink matrix). On average, the Mahalanobis distance between PASS–2nd-person videos and the remaining conditions was 1.62 ± 0.41 , compared with 0.87 ± 0.26 among the other conditions (for full results, see the supplementary data 2). This pattern demonstrates that PASS–2nd-person videos were rated in a consistently distinct manner, while the remaining conditions clustered more tightly with only graded variability. Thus, Engagement Ratings in the group resulted in a differentiation between actions expressing potential direct involvement in interaction (PASS–2nd-person) from all other actions.

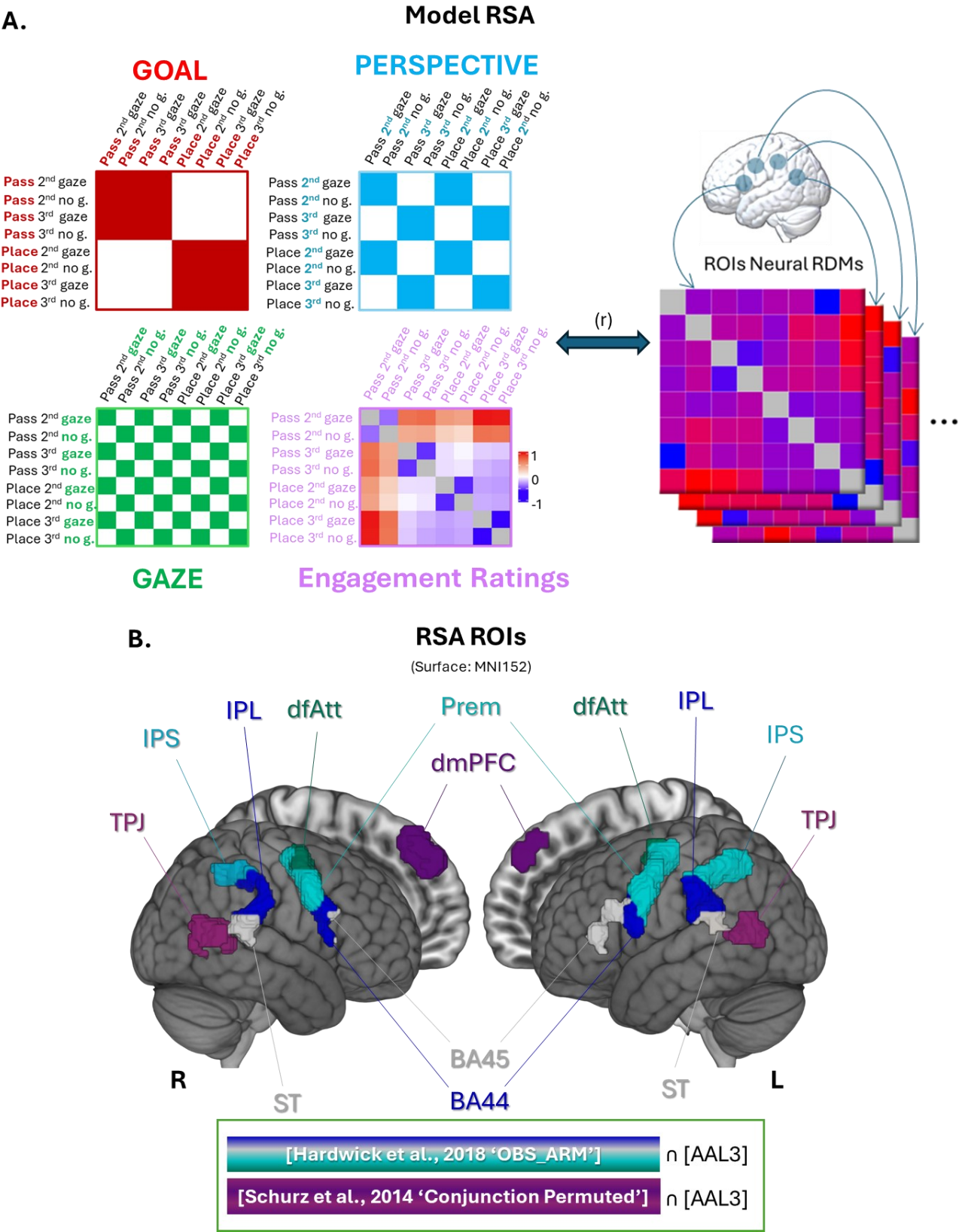
Second-Level RSA.

We performed two types of second-level RSA analyses: a *Neural RSA* and a *Model RSA*. In the Neural RSA, we computed pairwise distances between all ROI neural RDMs to evaluate inter-ROI representational similarity across conditions. In a Model RSA, each ROI neural RDM was compared with rating and conceptual RDMs to assess functional profiles. In both cases, Pearson correlations were calculated on the lower triangular portions of the RDMs (excluding the diagonal) and Fisher z-transformed before group-level inference (Fisher, 1915; Nili et al., 2014). The z-transformation was applied to correct for the skewed and bounded distribution of correlation coefficients, ensuring values were approximately normally distributed and suitable for parametric testing.

For the Neural RSA, subject-specific ROI RDMs were correlated, z-transformed, and entered into one-tailed t-tests across participants. False discovery rate (FDR) correction (Benjamini & Hochberg, 1995) was applied to control for multiple comparisons. To visualise representational relationships among ROIs, we applied hierarchical clustering and PCA to the average ROI RDMs. Clustering was implemented using the *hpc* function in *FactoMineR* (Lê et al., 2008), which builds a hierarchical tree via Ward's agglomerative method and identifies the optimal number of clusters by maximizing relative loss of inertia (Randriamihison et al., 2021). For visualisation purposes, a principal component analysis (PCA) was applied to project the clustering solution onto a two-dimensional factor map, with cluster labels derived from the hierarchical tree.

For the Model RSA, each ROI was tested against the four models. To evaluate both significant and null findings, we combined frequentist and Bayesian inference following the approach outlined by Keyser, Gazzola, and Wagenmakers (Keyser et al., 2020). First, one-tailed t-tests were run for each ROI–model pair ($\alpha = .05$, FDR-corrected). Second, Bayesian t-tests (JASP; Love et al., 2019) provided Bayes Factors (BF_{10}) to quantify graded evidence for the alternative (H_1) or null (H_0) hypotheses. Following Jeffreys' scale, Bayes Factors were interpreted continuously rather than dichotomously: values around 1 indicated anecdotal evidence; values ≥ 3 provided moderate evidence for H_1 ; and values ≥ 10 indicated strong evidence for H_1 . Because Bayes Factors are symmetric, values ≤ 0.33 indicated moderate evidence for H_0 , and values ≤ 0.1

430 indicated strong evidence for H_0 . Thus, higher BF_{10} values reflect stronger support for H_1 , whereas lower
 431 values (well below 1) reflect stronger support for H_0 . Results from both frequentist and Bayesian analyses
 432 are reported to jointly assess the strength of significant effects and the degree of evidence for absence.
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435 **Figure 2 | Representational models and ROIs.** (A) RDMs for the Goal model (red), Perspective model (light blue), Gaze
 436 model (green), and the mean Engagement model (purple frame), averaged among subjects' ratings. For Goal,
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Perspective, and Gaz, coloured cells represent condition pairs with 0 dissimilarity, and white cells represent maximum (1) dissimilarity. The 4 models were tested for correlation with RDMs computed from BOLD activity patterns (example on the right side of the figure) from each ROI for each participant. (B) Visualization of the selected ROIs on the MNI152 template.

ROI Selection.

To define our ROIs, we combined anatomical (atlas-based) and functional criteria, relying on meta-analytic data from fMRI studies on the AON and the MENTA network. We used the Automated Anatomical Labelling atlas 3 (AAL3; Rolls et al., 2020) for brain parcellation and applied this subdivision to binarised functional masks of the meta-analytic statistical maps, thereby assigning voxels to ROIs.

For the AON mask, we used the aggregated activation map for upper limb action observation from Hardwick et al. (Hardwick et al., 2018); available at <https://osf.io/3egd5>). This map includes voxels responsive to both action observation and execution, as well as those active during observation only. We decided to adopt this broader action observation map, considering the recently highlighted relevance of agent-specific responses in premotor cortices (cf. Bonini et al., 2022) along with ‘classic’ mirror (i.e., agent-shared) responses. Within this mask, we focused on lateral premotor, parietal, and temporal sectors, corresponding to core AON regions. For the MENTA mask, we used the conjunction map generated by permutation-based overlap across six meta-analyses of mentalizing tasks (Schurz et al., 2014; provided by the lead author). The FSL tool `fslmaths` was used to parcel these functional masks according to the AAL3 atlas (`aal_mni_v7_2mm.nii.gz`; <https://www.gin.cnrs.fr/en/tools/aal/>).

From the AON mask, 18 candidate ROIs were initially identified using AAL3 labels: Precentral L/R, Middle Frontal Gyrus L/R, Superior Frontal Gyrus L/R, IFG pars triangularis L/R, IFG pars opercularis L/R, Inferior Parietal Lobule L/R, Supramarginal Gyrus L/R, and Superior Temporal L/R. These were further refined based on anatomical and functional considerations. Precentral ROIs, overlapping mainly with BA6, were relabeled Premotor L/R. Dorsal frontal voxels from Middle and Superior Frontal Gyri extended rostral to BA6, likely overlapping with the dorsal fronto-parietal attention network (Corbetta & Shulman, 2002). Overlap with NeuroSynth attention maps confirmed this, so these areas were merged into dorsofrontal Attention ROIs (`dfAtt L/R`). Inferior Parietal ROIs were relabeled as IntraParietal Sulcus L/R, as they bordered somatosensory cortex and sat above Supramarginal Gyri, which were relabeled Inferior Parietal L/R. IFG ROIs were renamed BA44 L/R and BA45 L/R, based on their overlap with Brodmann areas. Superior Temporal ROIs were labelled ST L/R. This resulted in 14 ROIs from the AON mask (Figure 2B).

From the MENTA mask, 4 ROIs were defined: Superior Medial Frontal L/R (renamed dorsomesial Prefrontal, `dmPFC L/R`) and Temporal Medial L/R (renamed Temporo-parietal Junction, `TPJ L/R`).

In total, 18 ROIs were used for the RSA (Figure 2B). All first- and second-level RSAs were conducted with custom R scripts.

Noise Ceiling estimation.

To assess the reliability of the neural representational dissimilarity matrices (RDMs) and to contextualise the magnitude of RSA model fits, we computed the noise ceiling, an estimate of the maximum explainable variance that any model could achieve given inter-subject variability (Nili et al., 2014). Noise ceilings were calculated using a leave-one-subject-out approach. For the upper bound, each subject's neural RDM was correlated with the group average RDM (including the subject). For the lower bound, each subject's RDM was correlated with the group average computed excluding that subject. Correlations were computed using the Pearson correlation on the vectorised lower triangle of the neural RDMs.

We calculated both a global noise ceiling (averaged across the entire set of ROI $\times$ ROI distances) and per-ROI noise ceilings, where the analysis was restricted to each ROI's pattern of similarity with all other ROIs.

3. RSA Results

Noise ceiling. Noise ceiling estimates on ROIs' neural RDMs revealed consistent reliability of representational geometry across subjects, with upper bounds ranging from $r = 0.644$ (TPJ L) to $r = 0.773$ (dmPFC R). Most regions, including premotor, inferior parietal, and frontal ROIs, showed lower bounds above $r \approx 0.65$, while areas such as dmPFC reached > 0.70 , indicating robust inter-subject consistency in representational structure. Global noise ceilings were $r = 0.481$ (upper) and $r = 0.417$ (lower), reflecting substantial explainable variance in neural RDMs across ROIs. Full results and discussion are reported in the Supplementary Material (page 7).

Neural RSA. Figure 3 presents the patterns of correlations between neural RDMs, displayed both as a correlation matrix (Figure 3A) and as topological connectivity (Figure 3C), along with the clustering of similarity patterns among ROIs (Figure 3B).

The RSA revealed three key features of neural similarity across AON and MENTA regions:

- 1) AON frontoparietal ROIs and the TPJ exhibited extensive representational similarity, most pronounced in the rTPJ.
- 2) The dmPFC showed the greatest representational distance from other ROIs, with significant correlations restricted to its contralateral homologue.
- 3) Inter-hemispheric similarities were largely intra-network (i.e., AON-AON), with the only cross-network exception being the rTPJ and the left dorsofrontal Attention ROI. Inter-hemispheric correlations were primarily observed in midline and dorsal regions, including the bilateral IPS, dorsofrontal Attention ROIs, and the left premotor cortex.

The cluster analysis (Figure 3B) revealed six similarity clusters. Two were left-lateralized, one linking the left IPL with the Superior Temporal ROI (yellow), and another combining the left TPJ with BA45 (green). A broad, inter-hemispheric parieto-frontal cluster (cyan) joined the right dfAtt and IPS with the left IPS, BA44, and premotor area. Two predominantly right-lateralized clusters emerged, with one cluster grouping the ST, IPL, and BA45 (indigo), and another bringing together the TPJ, premotor area, and BA44 with the left dfAtt (pink). Finally, the dmPFC formed an isolated cluster (red). Visualization through dimensionality reduction further emphasized intra-hemispheric neural similarity, with left ROIs mostly occupying the upper part and the right ROIs mostly occupying the lower part of the factor map.

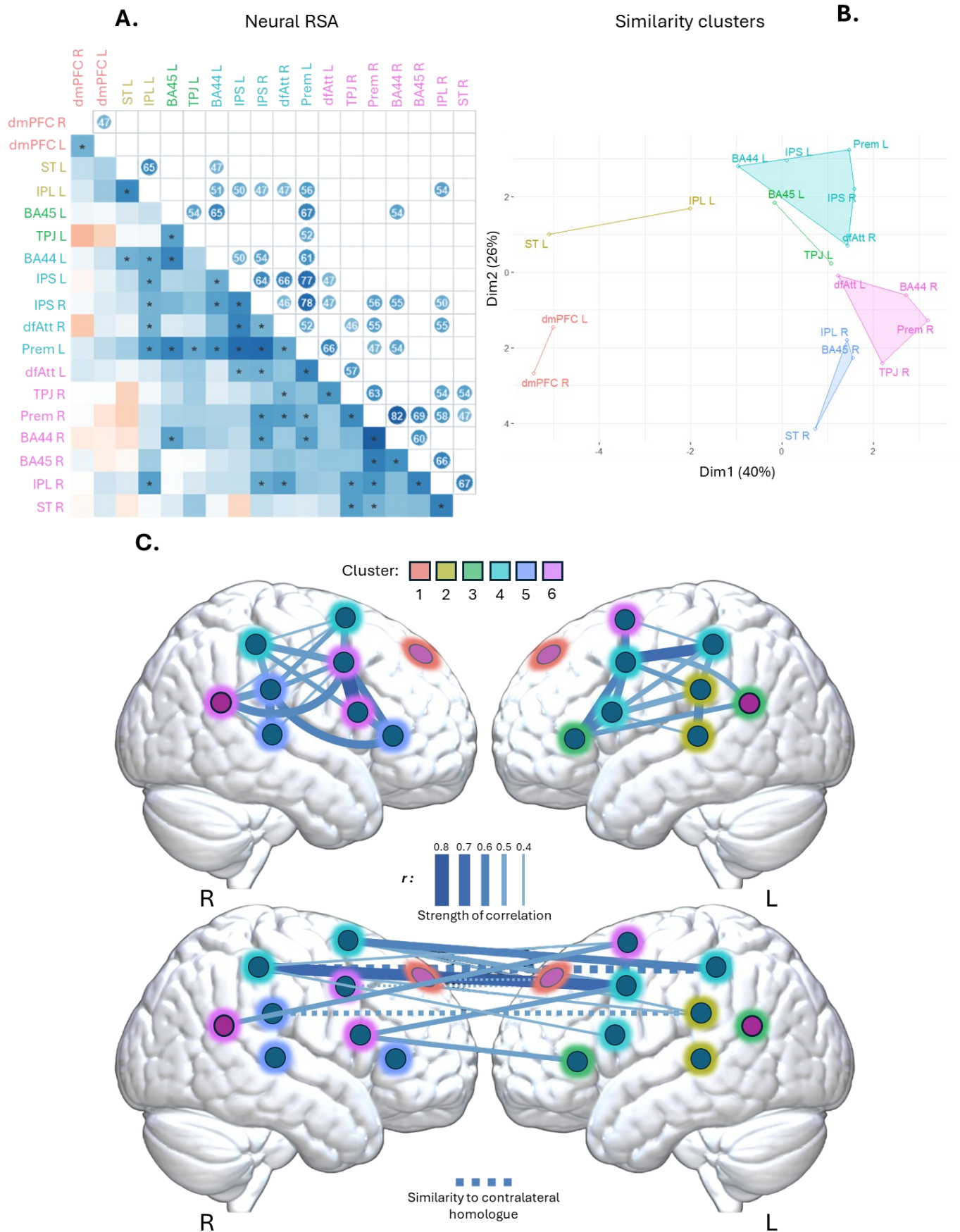


Figure 3| Neural RSA. (A) Dissimilarity matrix representing distance among ROIs' RDMs. The lower triangle highlights significant correlations (*), while the upper triangle reports the strength of significant correlations. (B) Plot of the

529 dimensionality reduction analysis with clusters obtained by the Unsupervised Hierarchical Clustering on the neural RSA.
530 (C) Correlations among ROI's RDMs are represented as topological connectivity plots: the upper one reporting intra-
531 hemisphere significant correlations; the lower one reporting inter-hemisphere significant correlations. Significant
532 correlation between homologue contralateral areas is represented with dashed lines. The colour of the glows around
533 the circles indicates the ROI's cluster.

534
535 Among frontal areas, the left Premotor ROI showed the strongest representational similarity with the
536 bilateral IPS, forming a cluster that also included the right dorsofrontal Attention ROI. It displayed the
537 broadest representational connectivity overall, spanning both intra- and inter-hemispheric correlations. By
538 contrast, the right Premotor ROI exhibited a more lateralized pattern, with only moderate similarity to its
539 contralateral homologue. It was significantly correlated with all frontoparietal and temporal ROIs, showing
540 its strongest similarity with BA44/45 and the TPJ.

541
542 Inferior frontal regions (BA44 and BA45) were most similar to adjacent ipsilateral Premotor ROIs. BA44's
543 significant similarities were largely confined to ipsilateral frontoparietal AON regions, whereas BA45 showed
544 hemispheric asymmetry: left BA45, but not right BA45, correlated with the ipsilateral TPJ.

545
546 The dorsofrontal Attention ROIs showed significant correlations both within and across hemispheres,
547 particularly with the ipsilateral Premotor ROI and contralateral IPS. Notably, the left dfAtt correlated with the
548 right TPJ, forming a cluster with right frontal regions (right Premotor and right BA44).

549
550 In temporal regions, Superior Temporal ROIs and the TPJs showed distinct patterns. The Superior Temporal
551 ROIs formed more localised, lateralized networks, showing the strongest similarity to the ipsilateral IPL and
552 at least one frontal ROI (left BA44 or right Premotor). IPLs also displayed primarily lateralized similarity, with
553 minimal inter-hemispheric correlations. In contrast, the TPJ exhibited broader representational similarity,
554 with differences between the left and right TPJ. The left TPJ correlated with ipsilateral frontal areas,
555 specifically the Premotor ROI and BA45, clustering with the latter. The right TPJ showed the strongest
556 similarity with the ipsilateral Premotor ROI, followed by the right IPL and the left dfAtt.

557
558 Finally, the dmPFC showed no significant correlations with any other ROI apart from its contralateral
559 counterpart, forming a highly isolated cluster.

560
561 **Model RSA.** Figure 4 shows two brain maps with ROIs coloured according to the model which yielded the
562 highest correlation (fig. 4A) and the strength of evidence for the correlation (fig. 4B). Heatmaps reporting the
563 statistical results of the model RSA is displayed beneath, with ROIs (rows) sorted following clusters obtained
564 from the neural RSA.

565
566 The '*Perspective*' model yielded the strongest and most consistent correlations, with robust Bayesian
567 evidence across both MENTA and AON regions. In particular, the right TPJ, left premotor area, and left
568 dorsofrontal Attention ROI showed the most reliable correspondence with this model.

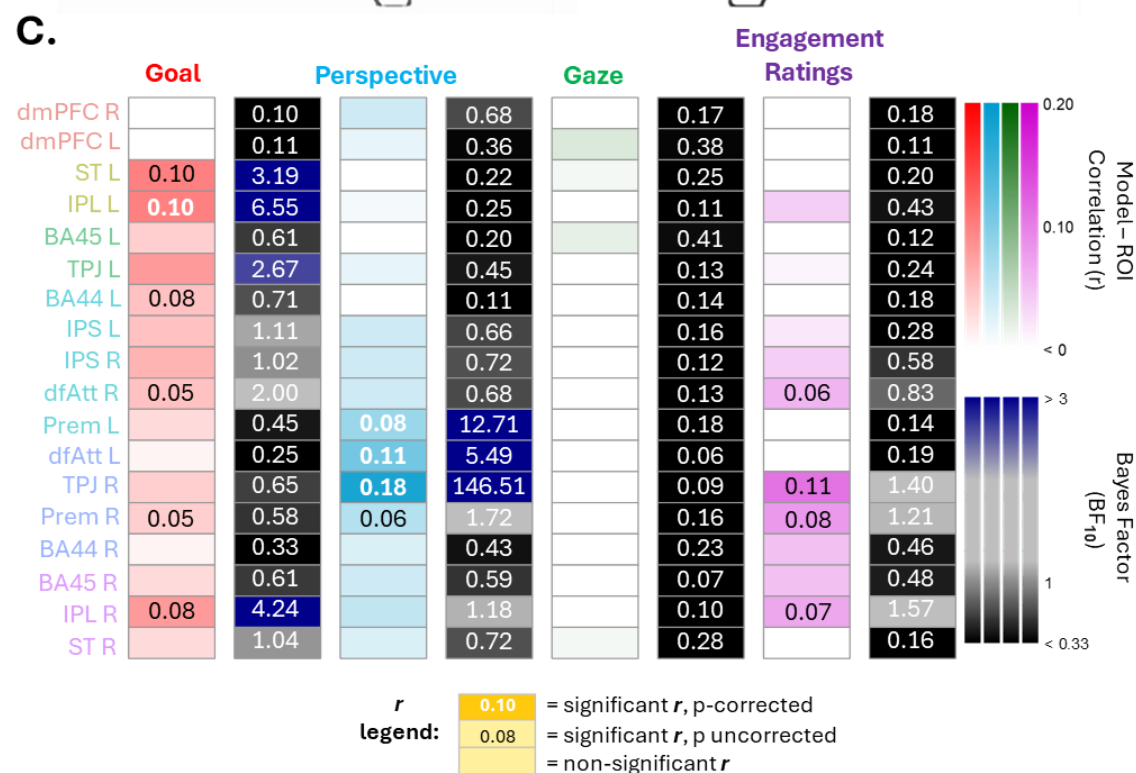
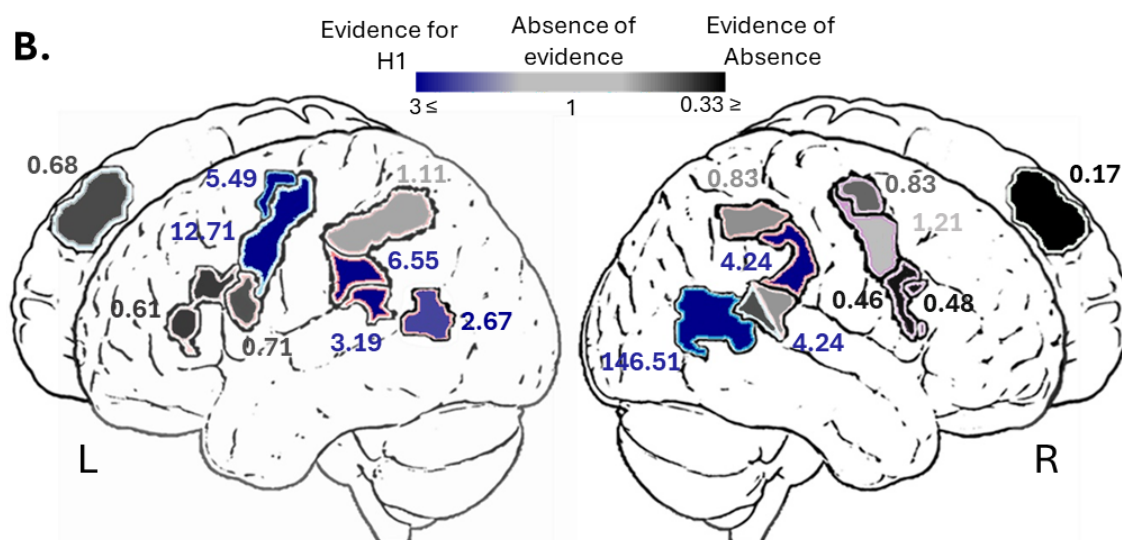
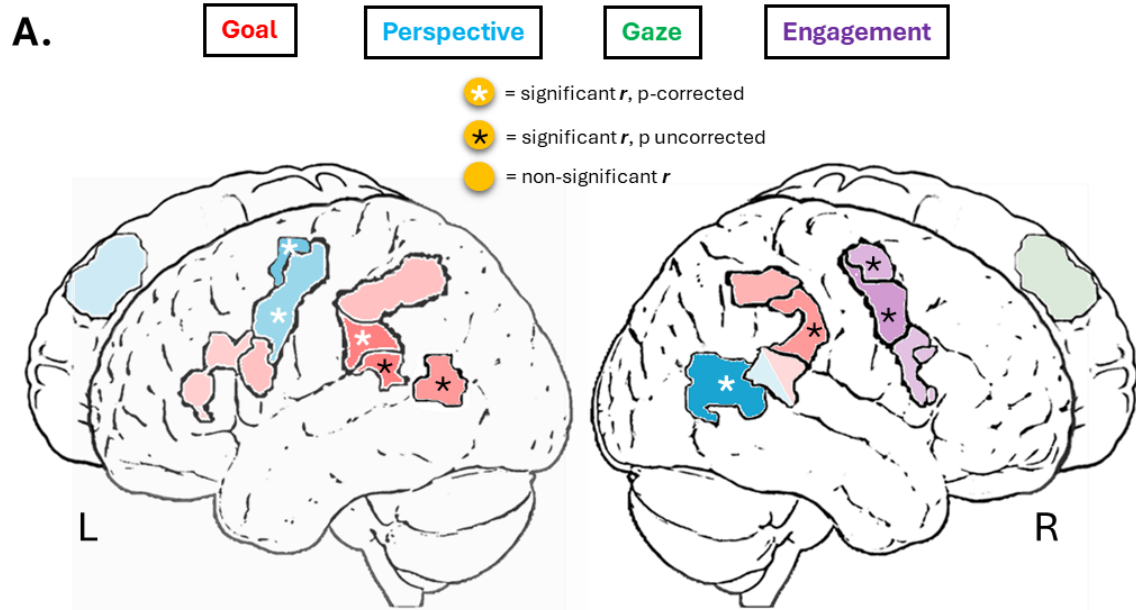


Figure 4 | Model RSA results. (A) Brain map with ROIs coloured according to the model which yielded the highest correlation score for that area. (B) Brain map with ROIs coloured based on the degree of evidence as measured by the Bayesian Factor. representing dissimilarity scores for the Model RSA. (C) Heatbars color-coded according to the strength of correlation (red, blue, green, and pink heatbars) and strength of evidence (BF_{10}), black-blue heatbars) of each model per ROI. Clusters from the neural RSA were employed to sort the ROIs to display how the neural similarity was predictive of functional similarity as indexed by the model RSA.

The ‘Goal’ model showed the strongest correlations and trends with left-lateralized, posterior ROIs, that is the left IPL, TPJ, and the Superior Temporal area. Bayesian analysis also highlighted evidence for a significant correlation between the ‘Goal’ model and the right IPL.

The ‘Gaze’ model yielded no significant results, indicating that gaze visibility was not a major factor affecting neural patterns during the task. Weak correlations in the left dmPFC and left Superior Temporal area were accompanied by strong Bayesian evidence for absence, suggesting that gaze visibility did not shape representational patterns in these ROIs.

Finally, the ‘Engagement Rating’ model also had no significant results. However, differently from the ‘Gaze’ model, it yielded high correlation values and statistical trends in a right-lateralized cluster of areas which included the right TPJ, right IPL, right Premotor ROI, and the right dfAtt. Interestingly, these areas also display proximity in terms of similarity patterns as highlighted by the clustering analysis (see Figure 4C, pink heatbar, and Figure 5, right bottom corner). Nonetheless, Bayesian evidence for the Engagement model was anecdotal in these ROIs (≈ 1).

Mapping of Model RSA to Neural RSA. Figure 5 recaps the relation between neural and model RSA by mapping each ROI’s strongest model correlation onto the Similarity clusters map. This visualisation highlights a Goal-based organization in the left hemisphere and a possible Engagement-based organization in the right hemisphere. Superimposed on this lateralized pattern, the right TPJ, left premotor ROI, and left dfAtt consistently aligned with the Perspective model, pointing to its encompassing role across both hemispheres.

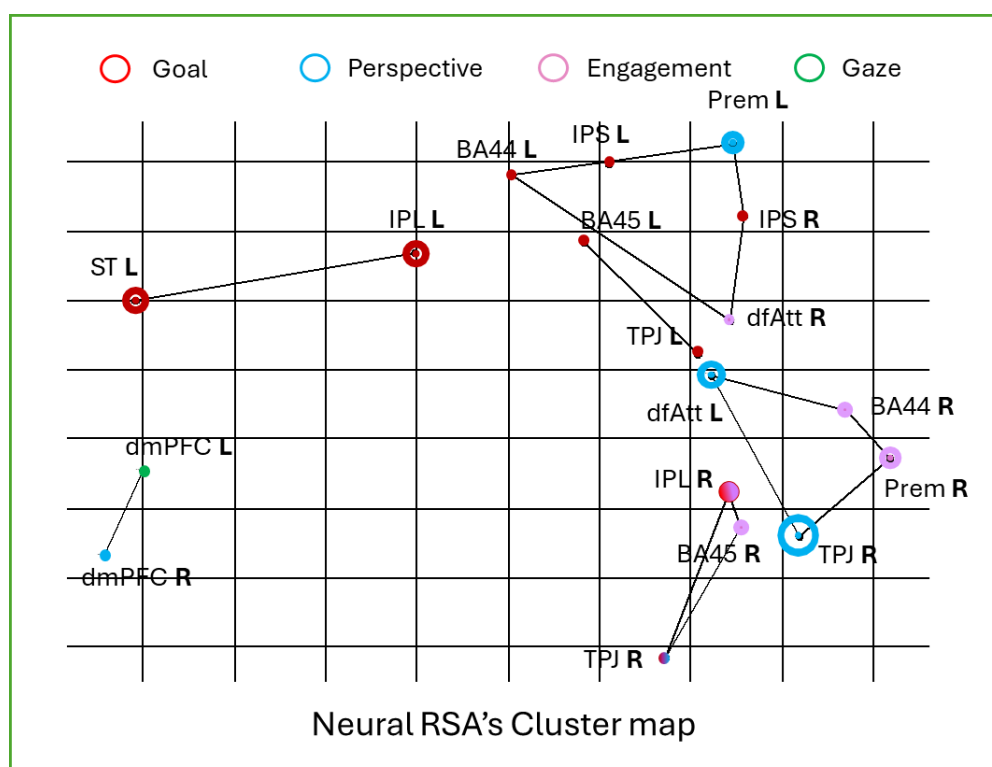


Figure 5 | Distribution of ROIs’ highest correlation to models on the neural RSA’s cluster map. The circles’ dimension correlates with the strength of the correlation.

624 3. Discussion

625

626 While much has been debated about how brain networks such as the AON and MENTA represent specific
627 features of others' behavior, less attention has been devoted to a crucial dimension of social stimuli: their
628 engaging nature for the observer. Building on previous research showing that potential engagement in
629 interpersonal interaction impacts how social information is processed in the brain, we conducted an fMRI
630 study examining cerebral responses during passive observation of the same action stimulus (object grasping)
631 presented with different combinations of engagement-relevant cues — Goal, Perspective, and Gaze —
632 designed to convey varying levels of potential social engagement.

633

634 Using multivariate RSA, we provide new evidence on how these networks jointly contribute to the neural
635 processing of actions affording potential interaction. Specifically, we assessed: 1) whether there is
636 representational connectivity across regions during action observation; 2) whether subregions show feature-
637 specific representational profiles consistent with conceptual models derived from previous literature; and 3)
638 whether engagement emerges as a distinct representational dimension.

639

640 4.1 Representational connectivity in the AON and MENTA: diverging pattern modulation in the dmPFC and 641 TPJ links the latter to premotor regions during action observation.

642

643 Our first aim was to test whether regions of the AON and MENTA systems show shared representational
644 structures during action observation. This was the goal of our Neural RSA (fig. 3), which assessed the extent
645 to which AON and MENTA's ROIs display a coordinated change in BOLD patterns in response to action stimuli
646 expressing varying levels of engagement. In this regard, our main finding is a striking dissociation between
647 the TPJ and the dmPFC, two regions often considered together as core "mentalizing" areas. In particular, the
648 TPJ showed robust and widespread representational similarity with frontal AON regions, whereas the dmPFC
649 appeared functionally isolated.

650

651 The left TPJ selectively correlated with the ipsilateral premotor cortex and BA45, forming a fronto-temporal
652 cluster. The rTPJ displayed instead a broader integration, correlating with the ipsilateral premotor cortex, IPL,
653 and superior temporal regions, as well as bilateral dorsofrontal Attention ROIs. This representational
654 connectivity profile combines with previous evidence hinting at the TPJ as a hub connecting sensorimotor
655 and attentional systems involved in action perception and control (Patel et al., 2019; Simos et al., 2017;
656 Szczepanski et al., 2013). In particular, our results add further evidence to the complementarity of TPJ and
657 premotor networks. Previous evidence shows TPJ involvement in motor execution, observation, and imagery
658 (Simos et al., 2017; Bachen li et al 2022), as well as causal modulation of motor responses during action
659 observation via transcranial stimulation (Giardina et al., 2011; Rauchbauer et al., 2015). Moreover, causal
660 interaction between the TPJ and primary motor/premotor cortices has been demonstrated in the regulation
661 of motor responses to aversive stimuli (Sagliano et al., 2019) and suppression of interference effects during
662 action observation (Bardi et al., 2017). Our findings extend this by showing TPJ-premotor representational
663 coupling to action stimuli even in passive action observation.

664

665 By contrast, the dmPFC stood apart from the rest of the network. Neither clustering nor model RSA revealed
666 significant similarity with other regions, aside from modest proximity to the left ST region. This is noteworthy,
667 as both the dmPFC and STS have been implicated in gaze processing (Hoffman & Haxby, 2000; Marquardt et
668 al., 2017). Yet, in our task, the correlation was non-significant, and Bayesian testing provided evidence for
669 absence. The dmPFC's distinctiveness was reinforced by the fact that the ST region itself aligned more closely
670 with a Goal-based cluster centred around temporo-parietal regions. Our finding that the dmPFC stood apart
671 from the rest of the network represents an interesting and unexpected negative result. Previous literature

consistently highlights dmPFC engagement with premotor and temporo-parietal structures in interactive or second-person contexts (Ciaramidaro et al., 2024; Saporta et al., 2022). Yet in our study, even though the stimuli combined engagement-relevant features affording or not affording interaction, no significant similarity with other regions emerged. A likely explanation is that while the task captured graded affordances of engagement, passive observation alone may not be sufficient to recruit the dmPFC. One possibility is that involvement of dmPFC fundamentally depends on sustained joint attention (Luft et al., 2022) during reciprocal social exchange. Another, not mutually exclusive explanation, is that its involvement is also highly dependent on task demands, as highlighted by its involvement during strategy selection (rather than other agents' beliefs representation) (Konovalov et al., 2021; Mahmoodi et al., 2023) and others' behavior tracking during joint interactions (Ninomiya et al., 2020; Yoshida et al., 2012). Thus, its contribution may require richer interactive demands, with reciprocal exchanges, consistent with the view that the dmPFC plays a role in managing ongoing interactions rather than processing isolated action features.

4.2 Feature specificity: action direction as a major modulator of neural activity during action observation.

While MENTA regions showed markedly different similarity profiles, our Neural RSA confirmed robust intra-network similarity among frontoparietal regions of the AON, including the Premotor cortex, BA44, IPL, and IPS, as expected from their well-established anatomo-functional organisation (Bruni et al., 2017; Mendoza & Merchant, 2014). Yet, both the Neural RSA and the Model RSA — which assessed the functional profiles of ROIs' representational geometry — also revealed that, together with the Superior Temporal areas, TPJs, and dorsofrontal Attention regions, these areas contributed to clusters reflecting distinct functional trends.

The bilateral IPLs showed strong similarity to ipsilateral ST regions and aligned with the Goal model, consistent with their known role in encoding action goals (Borra et al., 2008; Kilintari et al., 2014; Perrett et al., 1989). Goal-based coding was particularly strong in the left IPL and ST, with additional evidence in the left TPJ and right IPL. This finding aligns with works employing intracortical and causal techniques showing parietal cortex involvement in representing action classes (Aflalo et al., 2020; Lanzilotto et al., 2020; Urgen & Orban, 2021), and adds support to the view that perceptual processing of biological motion in ST is tightly coupled with goal representation in IPL.

Evidence for goal-coding in the ITPJ aligns with a recent RSA showing that intentions from motion kinematics and not belief-reasoning better explains ITPJ patterns across multiple mentalizing tasks (Golec-Staśkiewicz et al., 2022). Interestingly, however, the left TPJ did not cluster with the left ST and IPL. This may suggest that, although engaged in social cue processing — possibly due to shared visual inputs with the ST (Pitcher & Ungerleider, 2021) — the ITPJ and ST-IPL employ action-related information differently. The TPJ's precise role in sensorimotor transformation remains debated, but its established involvement in embodiment processes (Pryke et al., 2025) and multimodal integration suggests a function in managing internal models of self- and other-related actions, as further evidenced by its role in resolving conflicting self-other representations (Arioli et al., 2023; Quesque & Brass, 2019). Supporting this view, decoding of left TPJ patterns discriminates between self and other actions in working memory (Woźniak et al., 2022). Our results provide indirect evidence consistent with this hypothesis.

The strongest evidence for engagement-feature encoding however, came from the Perspective model, which correlated significantly with the left premotor cortex, rTPJ, and left dfAtt. A Perspective-based representational structure in the premotor cortex aligns with prior RSA (Liu et al., 2025; Urgen et al., 2019) and univariate work (Begliomini et al., 2017; Centelles et al., 2011) highlighting premotor encoding of action

718 directed toward or away from the observer. Previous evidence for the relevance of direction encoding also
719 applies to the dfAtt ROI, with findings showing that dorsal premotor lesions, but not IPL, impair imitation of
720 movement trajectories (Wong et al., 2019). Premotor directional tuning, established in mirror neurons
721 (Gallese et al., 1996; Maranesi et al., 2024), is closely linked to action space (Bonini et al., 2014; Fogassi et al.,
722 1996), suggesting that observed actions are mapped as motor affordances onto the observer's motor system
723 (Bonini et al., 2023; Orban et al., 2021). In this regard, action direction might be a major social affordance
724 during action observation. Our univariate contrasts (see Supplementary Material) further support a
725 pragmatic interpretation of action processing: second-person perspectives engaged ventral occipitotemporal
726 cortex and opercular IFG, regions linked to kinematic extraction and hand-action control (Borra et al., 2017;
727 Peelen & Downing, 2005), whereas third-person perspectives recruited dorsal parietal and precuneus
728 regions, associated with spatial monitoring and perspective-taking (Corbetta & Shulman, 2011; Szczepanski
729 et al., 2013). Second-person perspectives thus seemed to enhance premotor and occipitotemporal
730 processing for interaction affordances, while third-person perspectives preferentially recruit dorsal
731 attentional systems.

732

733 Prima facie, the Perspective-based representational geometry in the rTPJ supports accounts linking this
734 region to perspective-taking (Martin et al., 2020; Schurz et al., 2013). Yet, our study cannot determine
735 whether this reflects low-level mechanisms such as reconstructing gaze direction or higher-level processes
736 like simulating a first-person perspective. Prior RSA studies have mostly examined abstract mental state
737 representations in the rTPJ during mentalizing tasks, showing that it encodes information about others'
738 emotions and causal beliefs (Corradi-Dell'Acqua et al., 2014; Koster-Hale et al., 2017). However, more
739 recently, Merchant et al. (Merchant et al., 2022) demonstrated that rTPJ representational patterns are better
740 captured by interaction-based models (e.g., interacting with a peer vs. an abstract character) rather than by
741 an active during mentalization vs. non-mentalization distinction.

742

743 In line with this, our findings suggest that the TPJ is not primarily devoted to the encoding of high-level
744 features. In our data, beyond perspective encoding, the rTPJ clustered with the right premotor cortex, and
745 together they showed a trend toward the Engagement Ratings model, which reflects sensitivity to first-
746 person interaction. Whereas Merchant et al. reported engagement discrimination in an interactive paradigm
747 explicitly requiring joint coordination, our results suggest that this can emerge during passive observation of
748 action stimuli, together with TPJ–premotor coupling. This suggests that the TPJ flexibly encodes interaction-
749 relevant information, regardless of whether active coordination is required.

750

751 **4.3 Engagement as a social affordance vs. engagement as a high-level feature.**

752

753 Beyond assessing representational connectivity and feature specificity, we also tested whether perceived
754 engagement could be mapped onto neural patterns by comparing participants' ratings with ROI RDMS. Our
755 Engagement Rating model, however, did not reach robust significance. The strongest (though unreliable)
756 correlations emerged in a right-lateralized cluster including the rTPJ, rIPL, right premotor cortex, and right
757 dfAtt. While no literature has directly linked right-hemisphere specialisation to second-person contexts,
758 right-lateralisation is well established for attentional control (Corbetta et al., 2008), and there is some
759 evidence correlating increased social interaction during speech production with stronger involvement of the
760 rTPJ (Alexandrou et al., 2017). Moreover, recent infant data showed that a right-lateralized temporo-frontal
761 network, more active during face-to-face than third-person interactions, predicts later social and linguistic
762 development (Bosseler et al., 2024). Our findings may align with this hemispheric bias, but the absence of

763 reliable RSA effects suggests that subjective engagement was not robustly captured at the representational
764 level in this paradigm.

765
766 Several factors may explain these findings, some regarding the nature of the task. First, while participants'
767 ratings clearly distinguished second-person social actions from other stimuli, our passive task, with repetitive
768 trials and no demand for online interaction, may have dampened engagement processing. Indeed, passive-
769 viewing studies similarly report weak dmPFC involvement in third-person action and interaction observation
770 (Lee Masson & Isik, 2021; Otti et al., 2015). Second, the lack of an explicit task may have made PASS and
771 PLACE actions similarly engaging in terms of the social affordance they represent: both involved grasping
772 after another's action, differing only in the object grasped (hand vs. cylinder). Supporting this, univariate
773 contrasts showed no significant effects for PASS > PLACE, but PLACE > PASS significantly activated bilateral
774 parietal operculum overlapping with SII. This likely reflects the physical contact of the object with the
775 cylinder, consistent with evidence that somatosensory cortices, particularly SII, are recruited during goal-
776 directed actions (Gazzola & Keysers, 2009; Keysers et al., 2010), carrying information regarding vicarious
777 touch (Aberbach-Goodman & Mukamel, 2023; Bellard et al., 2023) and physical causality (Karakose-Akbiyik
778 et al., 2023; Keysers et al., 2004).

779
780 Similarly, and contrary to our hypotheses, the Gaze model produced no significant correlations, a null result
781 further consolidated by Bayesian evidence for absence. This was mirrored in the univariate contrasts, where
782 the only effect observed was greater activation in the left lingual cortex for the No Gaze > Gaze contrast,
783 rather than in canonical gaze-processing regions such as the STs, Frontal Eye Fields, or the IPS. This finding is
784 striking, given previous RSA evidence for robust, invariant representations of gaze direction (Carlin et al.,
785 2011). One likely explanation is that in our paradigm, gaze played only a minimal role, as the only change in
786 gaze behavior through stimuli regarded the last ~500 ms of eye contact at the end of passing actions. Gaze
787 direction could be easily inferred from the action context, and repeated exposure to the same stimuli may
788 have reduced its saliency. Importantly, the relevance of gaze processing is not only determined by whether
789 gaze is direct or averted, but also by the broader interactional context. For instance, gaze has been shown to
790 differentially modulate neural activity depending on whether the situation is competitive or prosocial (Klein
791 et al., 2008; Sun et al., 2018). By contrast, our task involved passive viewing without interactive demands,
792 which may have further diminished the role of gaze. It is also possible that our instructions, which asked
793 participants to focus on the unfolding of the action rather than on specific features, biased attention away
794 from gaze. Without direct control of eye movements, however, this remains speculative.

795
796 Beyond task and methodological factors, another explanation for our negative result may lie in the nature of
797 engagement itself. Our results suggest that engagement is not a distinct neural dimension but instead
798 emerges from the integration of goal encoding and directional cues within a distributed sensorimotor
799 network including the rTPJ, IPL, premotor, and dorsofrontal Attention areas. In this view, engagement reflects
800 a property of combined features, such as social goals or the pragmatic meaning of spatio-temporal cues,
801 consistent with the social affordance hypothesis of action observation. Supporting this view, a recent RSA
802 study on animacy (Jozwik et al., 2022) showed that animacy judgments mapped onto neural patterns only
803 when decomposed into finer dimensions, such as "having agency" or "having mobility." Additional factors
804 likely shape engagement, such as reciprocity, contextual expectations, social saliency, or the relationship with
805 the observed agent — variables not manipulated in our experiment but that could further map onto AON
806 and MENTA regions during online interactions.

807
808

809 5. Outlook and limitations

810

811 Our study has several limitations that also point toward promising directions for future research. First, the
812 task was passive and non-interactive, likely under-recruiting regions such as the dmPFC and the BA44-BA45
813 composing the Inferior Frontal Gyrus (IFG), which are especially sensitive to second-person contexts (Redcay
814 & Schilbach, 2019). As discussed, task instructions emphasised action goals, potentially biasing
815 representational profiles, and the high stimulus repetition may have reduced sensitivity to subtle
816 modulations of engagement features. Future work should employ interactive, second-person paradigms,
817 manipulate attentional focus, and use more varied, naturalistic stimuli to capture engagement
818 representations beyond simple direction and gaze cues.

819

820 At the neural level, our findings emphasise robust TPJ–premotor representational connectivity but leave
821 open the exact mechanisms by which the TPJ exerts synergistic actions with parieto-frontal motor regions.
822 Future RSA studies further manipulating self–other distinction and perspective-taking, particularly in second-
823 versus third-person contexts, may clarify TPJ’s functional profile. Moreover, combining RSA with dynamic
824 connectivity analyses could further track how feature-specific codes propagate.

825

826 In contrast, our analyses did not confirm the similarity between the dmPFC and any frontal ROIs,
827 underscoring its distinct role. However, in light of the previous literature, our negative result might be a piece
828 of further evidence for dmPFC playing an eminently executive control structure within the social brain,
829 particularly sensitive to the on-demand management of multifaceted social information in social tasks
830 ranging from online motor interactions (Báez-Mendoza et al., 2021; Ninomiya et al., 2020) to self-other
831 comparison (Wittmann et al., 2021). Moreover, the dmPFC may encode broader dynamics of interaction
832 sequences, requiring models or tasks that capture representational dimensions beyond those tested here,
833 like action/interaction sequences or strategy tracking.

834

835 6. Conclusions

836

837 Interpersonal motor interactions are a core part of daily social life, requiring rapid integration of information
838 about one’s own movements and those of others. Decades of research suggest that this process is not
839 segregated into distinct neural modules, but interwoven across regions for bodily control and goal-directed
840 action, creating a “we-centric” neural space (Gallese, 2006; Gallotti & Frith, 2013) where agent-related
841 information about the self and others is mapped and compared.

842

843 Our RSA contributes to this picture by showing coordinated changes in neural patterns between premotor
844 regions of the AON and the TPJ during action observation, linking these patterns to the encoding of
845 engagement-relevant features. The TPJ, together with frontoparietal regions, formed representational
846 clusters that captured action goals and first-person interaction affordances, as signalled by action direction
847 and increasing perceived engagement. These findings extend evidence that involvement in interaction
848 recruits a broad, distributed network tuned to features critical for managing social exchanges.

849

850 By contrast, the dmPFC did not cluster with either TPJ or AON regions and showed no alignment with Goal
851 or Engagement models. Rather than undermining its role in social cognition, this emphasises its sensitivity to
852 task demands and reciprocity, consistent with accounts positioning the dmPFC as an executive hub recruited
853 when social exchanges require active coordination or higher-order control.

854

855 Taken together, these results argue against a strict dichotomy between abstract mentalizing and embodied
856 motor coding. Instead, they support recent views that social brain computations are best understood through

857 their pragmatic, embodied basis, providing a more dynamic account of how the brain encodes others'
858 behaviours in an interaction-oriented way.

859

860 **Additional Information.**

861

862 **Author contributions.**

863 E.A., M.A., and V.Gal. designed the experiment. E.A., F.F., and N.L. set up the experimental setting and
864 performed data acquisition. E.A., C.K., and V.Gaz. designed the analyses. E.A. and L.C. performed data
865 analyses. E.A., C.K., V.Gaz., and V.Gal. interpreted the results. E.A. wrote the original draft of the paper. C.K.,
866 V.Gaz., and V.Gal. reviewed and edited the paper. All authors have contributed to, seen, and approved the
867 manuscript.

868

869 **Data and Code availability.**

870 All additional materials, data, and custom analysis code supporting the findings of this study are available
871 from the corresponding author upon reasonable request.

872

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877

878 **Competing interests.**

879 The authors declare no competing interests.

880

881 **References.**

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883 Abe, M. O., Koike, T., Okazaki, S., Sugawara, S. K., Takahashi, K., Watanabe, K., & Sadato, N.
884 (2019). Neural correlates of online cooperation during joint force production. *NeuroImage*,
885 191, 150–161. <https://doi.org/10.1016/j.neuroimage.2019.02.003>

886 Aberbach-Goodman, S., & Mukamel, R. (2023). Temporal hierarchy of observed goal-directed
887 actions. *Scientific Reports*, 13(1), 19701. <https://doi.org/10.1038/s41598-023-46917-z>

888 Aflalo, T., Zhang, C. Y., Rosario, E. R., Pouratian, N., Orban, G. A., & Andersen, R. A. (2020). A
889 shared neural substrate for action verbs and observed actions in human posterior parietal
890 cortex. *Science Advances*, 6(43), eabb3984. <https://doi.org/10.1126/sciadv.abb3984>

891 Alcalá-López, D., Vogeley, K., Binkofski, F., & Bzdok, D. (2019). Building blocks of social
892 cognition: Mirror, mentalize, share? *Cortex*, 118, 4–18.
893 <https://doi.org/10.1016/j.cortex.2018.05.006>

894 Alexandrou, A. M., Saarinen, T., Mäkelä, S., Kujala, J., & Salmelin, R. (2017). The right
895 hemisphere is highlighted in connected natural speech production and perception.
896 *NeuroImage*, 152, 628–638. <https://doi.org/10.1016/j.neuroimage.2017.03.006>

897 Angelini, M., Del Vecchio, M., Lopomo, N. F., Gobbo, M., & Avanzini, P. (2021). Perspective-
898 dependent activation of frontoparietal circuits during the observation of a static body
899 effector. *Brain Research*, 1769, 147604. <https://doi.org/10.1016/j.brainres.2021.147604>

900 Arcuri, E., Ardizzi, M., & Gallese, V. (2025). Reading out bodily cues to predict interactions.
901 *Scientific Reports*, 15(1), 18865. <https://doi.org/10.1038/s41598-025-01224-7>

902 Arioli, M., Cattaneo, Z., Parimbelli, S., & Canessa, N. (2023). Relational vs representational social
903 cognitive processing: A coordinate-based meta-analysis of neuroimaging data. *Social*
904 *Cognitive and Affective Neuroscience*, 18(1), nsad003. <https://doi.org/10.1093/scan/nsad003>

- 905 Báez-Mendoza, R., Mastrobattista, E. P., Wang, A. J., & Williams, Z. M. (2021). Social agent
906 identity cells in the prefrontal cortex of interacting groups of primates. *Science*, 374(6566),
907 eabb4149. <https://doi.org/10.1126/science.abb4149>
- 908 Bardi, L., Six, P., & Brass, M. (2017). Repetitive TMS of the temporo-parietal junction disrupts
909 participant's expectations in a spontaneous Theory of Mind task. *Social Cognitive and*
910 *Affective Neuroscience*, 12(11), 1775–1782. <https://doi.org/10.1093/scan/nsx109>
- 911 Becchio, C., Cavallo, A., Begliomini, C., Sartori, L., Feltrin, G., & Castiello, U. (2012). Social
912 grasping: From mirroring to mentalizing. *NeuroImage*, 61(1), 240–248.
913 <https://doi.org/10.1016/j.neuroimage.2012.03.013>
- 914 Begliomini, C., Cavallo, A., Manera, V., Becchio, C., Stramare, R., Miotto, D., & Castiello, U.
915 (2017). Potential for social involvement modulates activity within the mirror and the
916 mentalizing systems. *Scientific Reports*, 7(1), 14967. [https://doi.org/10.1038/s41598-017-](https://doi.org/10.1038/s41598-017-14476-9)
917 14476-9
- 918 Bekkali, S., Youssef, G. J., Donaldson, P. H., He, J., Do, M., Hyde, C., Barhoun, P., & Enticott, P.
919 G. (2022). Do gaze behaviours during action observation predict interpersonal motor
920 resonance? *Social Cognitive and Affective Neuroscience*, 17(1), 61–71.
921 <https://doi.org/10.1093/scan/nsaa106>
- 922 Bellard, A., Trotter, P. D., McGlone, F. L., & Cazzato, V. (2023). Role of medial prefrontal cortex
923 and primary somatosensory cortex in self and other-directed vicarious social touch: A TMS
924 study. *Social Cognitive and Affective Neuroscience*, 18(1), nsad060.
925 <https://doi.org/10.1093/scan/nsad060>
- 926 Benjamini, Y., & Hochberg, Y. (1995). *Controlling the False Discovery Rate: A Practical and*
927 *Powerful Approach to Multiple Testing*. 57, 289–300.
- 928 Blanke, O., & Arzy, S. (2005). The Out-of-Body Experience: Disturbed Self-Processing at the
929 Temporo-Parietal Junction. *The Neuroscientist*, 11(1), 16–24.
930 <https://doi.org/10.1177/1073858404270885>
- 931 Boccadoro, S., Cracco, E., Hudson, A. R., Bardi, L., Nijhof, A. D., Wiersema, J. R., Brass, M., &
932 Mueller, S. C. (2019). Defining the neural correlates of spontaneous theory of mind (ToM):
933 An fMRI multi-study investigation. *NeuroImage*, 203, 116193.
934 <https://doi.org/10.1016/j.neuroimage.2019.116193>
- 935 Bonini, L., Maranesi, M., Livi, A., Fogassi, L., & Rizzolatti, G. (2014). Space-dependent
936 representation of objects and other's action in monkey ventral premotor grasping neurons.
937 *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 34(11),
938 4108–4119. <https://doi.org/10.1523/JNEUROSCI.4187-13.2014>
- 939 Bonini, L., Rotunno, C., Arcuri, E., & Gallese, V. (2022). Mirror neurons 30 years later:
940 Implications and applications. *Trends in Cognitive Sciences*, 26(9), 767–781.
941 <https://doi.org/10.1016/j.tics.2022.06.003>
- 942 Bonini, L., Rotunno, C., Arcuri, E., & Gallese, V. (2023). The mirror mechanism: Linking
943 perception and social interaction. *Trends in Cognitive Sciences*, 27(3), 220–221.
944 <https://doi.org/10.1016/j.tics.2022.12.010>
- 945 Borra, E., Belmalih, A., Calzavara, R., Gerbella, M., Murata, A., Rozzi, S., & Luppino, G. (2008).
946 Cortical Connections of the Macaque Anterior Intraparietal (AIP) Area. *Cerebral Cortex*,
947 18(5), 1094–1111. <https://doi.org/10.1093/cercor/bhm146>
- 948 Borra, E., Gerbella, M., Rozzi, S., & Luppino, G. (2017). The macaque lateral grasping network: A
949 neural substrate for generating purposeful hand actions. *Neuroscience & Biobehavioral*
950 *Reviews*, 75, 65–90. <https://doi.org/10.1016/j.neubiorev.2017.01.017>
- 951 Bosseler, A. N., Meltzoff, A. N., Bierer, S., Huber, E., Mizrahi, J. C., Larson, E., Endevelt-Shapira,
952 Y., Taulu, S., & Kuhl, P. K. (2024). Infants' brain responses to social interaction predict
953 future language growth. *Current Biology: CB*, 34(8), 1731-1738.e3.
954 <https://doi.org/10.1016/j.cub.2024.03.020>

- 955 Bristow, D., Rees, G., & Frith, C. D. (2007). Social interaction modifies neural response to gaze
956 shifts. *Social Cognitive and Affective Neuroscience*, 2(1), 52–61.
957 <https://doi.org/10.1093/scan/nsl036>
- 958 Bruni, S., Gerbella, M., Bonini, L., Borra, E., Coudé, G., Ferrari, P. F., Fogassi, L., Maranesi, M.,
959 Rodà, F., Simone, L., Serventi, F. U., & Rozzi, S. (2017). Cortical and subcortical
960 connections of parietal and premotor nodes of the monkey hand mirror neuron network.
961 *Brain Structure and Function*. <https://doi.org/10.1007/s00429-017-1582-0>
- 962 Carlin, J. D., Calder, A. J., Kriegeskorte, N., Nili, H., & Rowe, J. B. (2011). A Head View-Invariant
963 Representation of Gaze Direction in Anterior Superior Temporal Sulcus. *Current Biology*,
964 21(21), 1817–1821. <https://doi.org/10.1016/j.cub.2011.09.025>
- 965 Cattaneo, L., Caruana, F., Jezzini, A., & Rizzolatti, G. (2009). Representation of Goal and
966 Movements without Overt Motor Behavior in the Human Motor Cortex: A Transcranial
967 Magnetic Stimulation Study. *The Journal of Neuroscience*, 29(36), 11134–11138.
968 <https://doi.org/10.1523/JNEUROSCI.2605-09.2009>
- 969 Cavallo, A., Lungu, O., Becchio, C., Ansuini, C., Rustichini, A., & Fadiga, L. (2015). When gaze
970 opens the channel for communication: Integrative role of IFG and MPFC. *NeuroImage*, 119,
971 63–69. <https://doi.org/10.1016/j.neuroimage.2015.06.025>
- 972 Centelles, L., Assaiante, C., Nazarian, B., Anton, J.-L., & Schmitz, C. (2011). Recruitment of Both
973 the Mirror and the Mentalizing Networks When Observing Social Interactions Depicted by
974 Point-Lights: A Neuroimaging Study. *PLoS ONE*, 6(1), e15749.
975 <https://doi.org/10.1371/journal.pone.0015749>
- 976 Cheng, X., Guo, B., & Hu, Y. (2022). Distinct neural couplings to shared goal and action
977 coordination in joint action: Evidence based on fNIRS hyperscanning. *Social Cognitive and*
978 *Affective Neuroscience*, 17(10), 956–964. <https://doi.org/10.1093/scan/nsac022>
- 979 Ciaramidaro, A., Becchio, C., Colle, L., Bara, B. G., & Walter, H. (2014). Do you mean me?
980 Communicative intentions recruit the mirror and the mentalizing system. *Social Cognitive*
981 *and Affective Neuroscience*, 9(7), 909–916. <https://doi.org/10.1093/scan/nst062>
- 982 Ciaramidaro, A., Toppi, J., Vogel, P., Freitag, C. M., Siniatchkin, M., & Astolfi, L. (2024). Synergy
983 of the mirror neuron system and the mentalizing system in a single brain and between brains
984 during joint actions. *NeuroImage*, 299, 120783.
985 <https://doi.org/10.1016/j.neuroimage.2024.120783>
- 986 Corbetta, M., Patel, G., & Shulman, G. L. (2008). The Reorienting System of the Human Brain:
987 From Environment to Theory of Mind. *Neuron*, 58(3), 306–324.
988 <https://doi.org/10.1016/j.neuron.2008.04.017>
- 989 Corbetta, M., & Shulman, G. L. (2002). Control of goal-directed and stimulus-driven attention in
990 the brain. *Nature Reviews. Neuroscience*, 3(3), 201–215. <https://doi.org/10.1038/nrn755>
- 991 Corradi-Dell’Acqua, C., Hofstetter, C., & Vuilleumier, P. (2014). Cognitive and affective theory of
992 mind share the same local patterns of activity in posterior temporal but not medial prefrontal
993 cortex. *Social Cognitive and Affective Neuroscience*, 9(8), 1175–1184.
994 <https://doi.org/10.1093/scan/nst097>
- 995 Coudé, G., Festante, F., Cilia, A., Loiacono, V., Bimbi, M., Fogassi, L., & Ferrari, P. F. (2016).
996 Mirror Neurons of Ventral Premotor Cortex Are Modulated by Social Cues Provided by
997 Others’ Gaze. *The Journal of Neuroscience*, 36(11), 3145–3156.
998 <https://doi.org/10.1523/JNEUROSCI.3220-15.2016>
- 999 Dal Monte, O., Fan, S., Fagan, N. A., Chu, C.-C. J., Zhou, M. B., Putnam, P. T., Nair, A. R., &
1000 Chang, S. W. C. (2022). Widespread implementations of interactive social gaze neurons in
1001 the primate prefrontal-amygdala networks. *Neuron*, 110(13), 2183–2197.e7.
1002 <https://doi.org/10.1016/j.neuron.2022.04.013>
- 1003 De Maesschalck, R., Jouan-Rimbaud, D., & Massart, D. L. (2000). The Mahalanobis distance.
1004 *Chemometrics and Intelligent Laboratory Systems*, 50(1), 1–18.
1005 [https://doi.org/10.1016/S0169-7439\(99\)00047-7](https://doi.org/10.1016/S0169-7439(99)00047-7)

- 1006 Del Vecchio, M., Avanzini, P., Gerbella, M., Costa, S., Zauli, F. M., d’Orio, P., Focacci, E., Sartori,
1007 I., & Caruana, F. (2024). Anato-mo-functional basis of emotional and motor resonance
1008 elicited by facial expressions. *Brain*, 147(9), 3018–3031.
1009 <https://doi.org/10.1093/brain/awae050>
- 1010 Ferrari-Toniolo, S., Visco-Comandini, F., & Battaglia-Mayer, A. (2019). Two brains in action: Joint-
1011 action coding in the primate frontal cortex. *The Journal of Neuroscience*, 1512–1518.
1012 <https://doi.org/10.1523/JNEUROSCI.1512-18.2019>
- 1013 Fisher, R. A. (1915). Frequency Distribution of the Values of the Correlation Coefficient in Samples
1014 from an Indefinitely Large Population. *Biometrika*, 10(4), 507.
1015 <https://doi.org/10.2307/2331838>
- 1016 Fogassi, L., Gallese, V., Fadiga, L., Luppino, G., Matelli, M., & Rizzolatti, G. (1996). Coding of
1017 peripersonal space in inferior premotor cortex (area F4). *Journal of Neurophysiology*, 76(1),
1018 141–157. <https://doi.org/10.1152/jn.1996.76.1.141>
- 1019 Frith, C. D., & Frith, U. (2021). Mapping Mentalising in the Brain. In M. Gilead & K. N. Ochsner
1020 (Eds.), *The Neural Basis of Mentalizing* (pp. 17–45). Springer International Publishing.
1021 https://doi.org/10.1007/978-3-030-51890-5_2
- 1022 Gallese, V. (2006). Intentional attunement: A neurophysiological perspective on social cognition
1023 and its disruption in autism. *Brain Research*, 1079(1), 15–24.
1024 <https://doi.org/10.1016/j.brainres.2006.01.054>
- 1025 Gallese, V. (2007). Before and below ‘theory of mind’: Embodied simulation and the neural
1026 correlates of social cognition. *Philosophical Transactions of the Royal Society B: Biological*
1027 *Sciences*, 362(1480), 659–669. <https://doi.org/10.1098/rstb.2006.2002>
- 1028 Gallese, V., Fadiga, L., Fogassi, L., & Rizzolatti, G. (1996). Action recognition in the premotor
1029 cortex. *Brain: A Journal of Neurology*, 119 (Pt 2), 593–609.
1030 <https://doi.org/10.1093/brain/119.2.593>
- 1031 Gallotti, M., & Frith, C. D. (2013). Social cognition in the we-mode. *Trends in Cognitive Sciences*,
1032 17(4), 160–165. <https://doi.org/10.1016/j.tics.2013.02.002>
- 1033 Gazzola, V., & Keysers, C. (2009). The observation and execution of actions share motor and
1034 somatosensory voxels in all tested subjects: Single-subject analyses of unsmoothed fMRI
1035 data. *Cerebral Cortex (New York, N.Y.: 1991)*, 19(6), 1239–1255.
1036 <https://doi.org/10.1093/cercor/bhn181>
- 1037 Geiger, A., Bente, G., Lammers, S., Tepest, R., Roth, D., Bzdok, D., & Vogeley, K. (2019). Distinct
1038 functional roles of the mirror neuron system and the mentalizing system. *NeuroImage*, 202,
1039 116102. <https://doi.org/10.1016/j.neuroimage.2019.116102>
- 1040 Giardina, A., Caltagirone, C., & Oliveri, M. (2011). Temporo-parietal junction is involved in
1041 attribution of hostile intentionality in social interactions: An rTMS study. *Neuroscience*
1042 *Letters*, 495(2), 150–154. <https://doi.org/10.1016/j.neulet.2011.03.059>
- 1043 Giese, M. A., & Rizzolatti, G. (2015). Neural and Computational Mechanisms of Action
1044 Processing: Interaction between Visual and Motor Representations. *Neuron*, 88(1), 167–180.
1045 <https://doi.org/10.1016/j.neuron.2015.09.040>
- 1046 Golec-Staśkiewicz, K., Pluta, A., Wojciechowski, J., Okruszek, Ł., Haman, M., Wysocka, J., &
1047 Wolak, T. (2022). Does the TPJ fit it all? Representational similarity analysis of different
1048 forms of mentalizing. *Social Neuroscience*, 17(5), 428–440.
1049 <https://doi.org/10.1080/17470919.2022.2138536>
- 1050 Guterstam, A., Bio, B. J., Wilterson, A. I., & Graziano, M. (2021). Temporo-parietal cortex involved
1051 in modeling one’s own and others’ attention. *eLife*, 10, e63551.
1052 <https://doi.org/10.7554/eLife.63551>
- 1053 Hardwick, R. M., Caspers, S., Eickhoff, S. B., & Swinnen, S. P. (2018). Neural correlates of action:
1054 Comparing meta-analyses of imagery, observation, and execution. *Neuroscience and*
1055 *Biobehavioral Reviews*, 94, 31–44. <https://doi.org/10.1016/j.neubiorev.2018.08.003>

- 1056 Haxby, J. V., Gobbini, M. I., Furey, M. L., Ishai, A., Schouten, J. L., & Pietrini, P. (2001).
 1057 Distributed and overlapping representations of faces and objects in ventral temporal cortex.
 1058 *Science (New York, N.Y.)*, 293(5539), 2425–2430. <https://doi.org/10.1126/science.1063736>
- 1059 Hayashi, T., Akikawa, R., Kawasaki, K., Egawa, J., Minamimoto, T., Kobayashi, K., Kato, S., Hori,
 1060 Y., Nagai, Y., Iijima, A., Someya, T., & Hasegawa, I. (2020). Macaques Exhibit Implicit
 1061 Gaze Bias Anticipating Others' False-Belief-Driven Actions via Medial Prefrontal Cortex.
 1062 *Cell Reports*, 30(13), 4433–4444.e5. <https://doi.org/10.1016/j.celrep.2020.03.013>
- 1063 Hoffman, E. A., & Haxby, J. V. (2000). Distinct representations of eye gaze and identity in the
 1064 distributed human neural system for face perception. *Nature Neuroscience*, 3(1), 80–84.
 1065 <https://doi.org/10.1038/71152>
- 1066 Hogeveen, J., Obhi, S. S., Banissy, M. J., Santiesteban, I., Press, C., Catmur, C., & Bird, G. (2015).
 1067 Task-dependent and distinct roles of the temporoparietal junction and inferior frontal cortex
 1068 in the control of imitation. *Social Cognitive and Affective Neuroscience*, 10(7), 1003–1009.
 1069 <https://doi.org/10.1093/scan/nsu148>
- 1070 Igelström, K. M., & Graziano, M. S. A. (2017). The inferior parietal lobule and temporoparietal
 1071 junction: A network perspective. *Neuropsychologia*, 105, 70–83.
 1072 <https://doi.org/10.1016/j.neuropsychologia.2017.01.001>
- 1073 Ionta, S., Heydrich, L., Lenggenhager, B., Mouthon, M., Fornari, E., Chapuis, D., Gassert, R., &
 1074 Blanke, O. (2011). Multisensory Mechanisms in Temporo-Parietal Cortex Support Self-
 1075 Location and First-Person Perspective. *Neuron*, 70(2), 363–374.
 1076 <https://doi.org/10.1016/j.neuron.2011.03.009>
- 1077 Jellema, T., Baker, C. I., Wicker, B., & Perrett, D. I. (2000). Neural representation for the perception
 1078 of the intentionality of actions. *Brain and Cognition*, 44(2), 280–302.
 1079 <https://doi.org/10.1006/breg.2000.1231>
- 1080 Jenkinson, M., Beckmann, C. F., Behrens, T. E. J., Woolrich, M. W., & Smith, S. M. (2012). FSL.
 1081 *NeuroImage*, 62(2), 782–790. <https://doi.org/10.1016/j.neuroimage.2011.09.015>
- 1082 Jozwik, K. M., Najarro, E., Van Den Bosch, J. J. F., Charest, I., Cichy, R. M., & Kriegeskorte, N.
 1083 (2022). Disentangling five dimensions of animacy in human brain and behaviour.
 1084 *Communications Biology*, 5(1), 1247. <https://doi.org/10.1038/s42003-022-04194-y>
- 1085 Kampe, K. K. W., Frith, C. D., & Frith, U. (2003). “Hey John”: Signals Conveying Communicative
 1086 Intention toward the Self Activate Brain Regions Associated with “Mentalizing,” Regardless
 1087 of Modality. *The Journal of Neuroscience*, 23(12), 5258–5263.
 1088 <https://doi.org/10.1523/JNEUROSCI.23-12-05258.2003>
- 1089 Karakose-Akbiyik, S., Caramazza, A., & Wurm, M. F. (2023). A shared neural code for the physics
 1090 of actions and object events. *Nature Communications*, 14(1), 3316.
 1091 <https://doi.org/10.1038/s41467-023-39062-8>
- 1092 Keysers, C., & Gazzola, V. (2007). Integrating simulation and theory of mind: From self to social
 1093 cognition. *Trends in Cognitive Sciences*, 11(5), Article 5.
 1094 <https://doi.org/10.1016/j.tics.2007.02.002>
- 1095 Keysers, C., Gazzola, V., & Wagenmakers, E.-J. (2020). Using Bayes factor hypothesis testing in
 1096 neuroscience to establish evidence of absence. *Nature Neuroscience*, 23(7), 788–799.
 1097 <https://doi.org/10.1038/s41593-020-0660-4>
- 1098 Keysers, C., Kaas, J. H., & Gazzola, V. (2010). Somatosensation in social perception. *Nature*
 1099 *Reviews Neuroscience*, 11(6), 417–428. <https://doi.org/10.1038/nrn2833>
- 1100 Keysers, C., Wicker, B., Gazzola, V., Anton, J.-L., Fogassi, L., & Gallese, V. (2004). A Touching
 1101 Sight. *Neuron*, 42(2), 335–346. [https://doi.org/10.1016/S0896-6273\(04\)00156-4](https://doi.org/10.1016/S0896-6273(04)00156-4)
- 1102 Kilintari, M., Raos, V., & Savaki, H. E. (2014). Involvement of the Superior Temporal Cortex in
 1103 Action Execution and Action Observation. *Journal of Neuroscience*, 34(27), 8999–9011.
 1104 <https://doi.org/10.1523/JNEUROSCI.0736-14.2014>

1105 Klein, J. T., Deaner, R. O., & Platt, M. L. (2008). Neural Correlates of Social Target Value in
1106 Macaque Parietal Cortex. *Current Biology*, 18(6), 419–424.
1107 <https://doi.org/10.1016/j.cub.2008.02.047>

1108 Kokal, I., Gazzola, V., & Keysers, C. (2009). Acting together in and beyond the mirror neuron
1109 system. *NeuroImage*, 47(4), 2046–2056. <https://doi.org/10.1016/j.neuroimage.2009.06.010>

1110 Kolling, N., & O'Reilly, J. X. (2018). State-change decisions and dorsomedial prefrontal cortex:
1111 The importance of time. *Current Opinion in Behavioral Sciences*, 22, 152–160.
1112 <https://doi.org/10.1016/j.cobeha.2018.06.017>

1113 Konovalov, A., Hill, C., Daunizeau, J., & Ruff, C. C. (2021). Dissecting functional contributions of
1114 the social brain to strategic behavior. *Neuron*, 109(20), 3323–3337.e5.
1115 <https://doi.org/10.1016/j.neuron.2021.07.025>

1116 Koster-Hale, J., Richardson, H., Velez, N., Asaba, M., Young, L., & Saxe, R. (2017). Mentalizing
1117 regions represent distributed, continuous, and abstract dimensions of others' beliefs.
1118 *NeuroImage*, 161, 9–18. <https://doi.org/10.1016/j.neuroimage.2017.08.026>

1119 Krall, S. C., Rottschy, C., Oberwelland, E., Bzdok, D., Fox, P. T., Eickhoff, S. B., Fink, G. R., &
1120 Konrad, K. (2015). The role of the right temporoparietal junction in attention and social
1121 interaction as revealed by ALE meta-analysis. *Brain Structure & Function*, 220(2), 587–604.
1122 <https://doi.org/10.1007/s00429-014-0803-z>

1123 Kriegeskorte, N. (2008). Representational similarity analysis – connecting the branches of systems
1124 neuroscience. *Frontiers in Systems Neuroscience*. <https://doi.org/10.3389/neuro.06.004.2008>

1125 Lanzilotto, M., Maranesi, M., Livi, A., Ferroni, C. G., Orban, G. A., & Bonini, L. (2020). Stable
1126 readout of observed actions from format-dependent activity of monkey's anterior
1127 intraparietal neurons. *Proceedings of the National Academy of Sciences*, 117(28), 16596–
1128 16605. <https://doi.org/10.1073/pnas.2007018117>

1129 Lê, S., Josse, J., & Husson, F. (2008). **FactoMineR**: An R Package for Multivariate Analysis.
1130 *Journal of Statistical Software*, 25(1). <https://doi.org/10.18637/jss.v025.i01>

1131 Lee Masson, H., & Isik, L. (2021). Functional selectivity for social interaction perception in the
1132 human superior temporal sulcus during natural viewing. *NeuroImage*, 245, 118741.
1133 <https://doi.org/10.1016/j.neuroimage.2021.118741>

1134 Lieberman, M. D., Straccia, M. A., Meyer, M. L., Du, M., & Tan, K. M. (2019). Social, self,
1135 (situational), and affective processes in medial prefrontal cortex (MPFC): Causal,
1136 multivariate, and reverse inference evidence. *Neuroscience & Biobehavioral Reviews*, 99,
1137 311–328. <https://doi.org/10.1016/j.neubiorev.2018.12.021>

1138 Lindquist, M. A., Meng Loh, J., Atlas, L. Y., & Wager, T. D. (2009). Modeling the hemodynamic
1139 response function in fMRI: Efficiency, bias and mis-modeling. *NeuroImage*, 45(1), S187–
1140 S198. <https://doi.org/10.1016/j.neuroimage.2008.10.065>

1141 Liu, S., Yu, L., Ren, J., Zhang, M., & Luo, W. (2025). The neural representation of body orientation
1142 and emotion from biological motion. *NeuroImage*, 310, 121163.
1143 <https://doi.org/10.1016/j.neuroimage.2025.121163>

1144 Luft, C. D. B., Zioga, I., Giannopoulos, A., Di Bona, G., Binetti, N., Civilini, A., Latora, V., &
1145 Mareschal, I. (2022). Social synchronization of brain activity increases during eye-contact.
1146 *Communications Biology*, 5(1), 412. <https://doi.org/10.1038/s42003-022-03352-6>

1147 Maffei, V., Indovina, I., Mazzarella, E., Giusti, M. A., Macaluso, E., Lacquaniti, F., & Viviani, P.
1148 (2020). Sensitivity of occipito-temporal cortex, premotor and Broca's areas to visible speech
1149 gestures in a familiar language. *PLOS ONE*, 15(6), e0234695.
1150 <https://doi.org/10.1371/journal.pone.0234695>

1151 Mahmoodi, A., Nili, H., Harbison, C., Hamilton, S., Trudel, N., Bang, D., & Rushworth, M. F. S.
1152 (2023). Causal role of a neural system for separating and selecting multidimensional social
1153 cognitive information. *Neuron*, 111(7), 1152–1164.e6.
1154 <https://doi.org/10.1016/j.neuron.2022.12.030>

1155 Maranesi, M., Lanzilotto, M., Arcuri, E., & Bonini, L. (2024). Mixed selectivity in monkey anterior
1156 intraparietal area during visual and motor processes. *Progress in Neurobiology*, 236, 102611.
1157 <https://doi.org/10.1016/j.pneurobio.2024.102611>

1158 Maranesi, M., Ugolotti Serventi, F., Bruni, S., Bimbi, M., Fogassi, L., & Bonini, L. (2013). Monkey
1159 gaze behaviour during action observation and its relationship to mirror neuron activity.
1160 *European Journal of Neuroscience*, 38(12), 3721–3730. <https://doi.org/10.1111/ejn.12376>

1161 Marquardt, K., Ramezanpour, H., Dicke, P. W., & Thier, P. (2017). Following Eye Gaze Activates a
1162 Patch in the Posterior Temporal Cortex That Is not Part of the Human “Face Patch” System.
1163 *ENEURO*, 4(2), ENEURO.0317-16.2017. <https://doi.org/10.1523/ENEURO.0317-16.2017>

1164 Martin, A. K., Kessler, K., Cooke, S., Huang, J., & Meinzer, M. (2020). The Right Temporoparietal
1165 Junction Is Causally Associated with Embodied Perspective-taking. *The Journal of*
1166 *Neuroscience*, 40(15), 3089–3095. <https://doi.org/10.1523/JNEUROSCI.2637-19.2020>

1167 McGettigan, C., Jasmin, K., Eisner, F., Agnew, Z. K., Josephs, O. J., Calder, A. J., Jessop, R.,
1168 Lawson, R. P., Spielmann, M., & Scott, S. K. (2017). You talkin’ to me? Communicative
1169 talker gaze activates left-lateralized superior temporal cortex during perception of degraded
1170 speech. *Neuropsychologia*, 100, 51–63.
1171 <https://doi.org/10.1016/j.neuropsychologia.2017.04.013>

1172 Mendoza, G., & Merchant, H. (2014). Motor system evolution and the emergence of high cognitive
1173 functions. *Progress in Neurobiology*, 122, 73–93.
1174 <https://doi.org/10.1016/j.pneurobio.2014.09.001>

1175 Merchant, J. S., Alkire, D., & Redcay, E. (2022). Neural similarity between mentalizing and live
1176 social interaction during the transition to adolescence. *Human Brain Mapping*, 43(13),
1177 4074–4090. <https://doi.org/10.1002/hbm.25903>

1178 Miss, F. M., Adriaense, J. E. C., & Burkart, J. M. (2022). Towards integrating joint action research:
1179 Developmental and evolutionary perspectives on co-representation. *Neuroscience &*
1180 *Biobehavioral Reviews*, 143, 104924. <https://doi.org/10.1016/j.neubiorev.2022.104924>

1181 Miyata, K., Koike, T., Nakagawa, E., Harada, T., Sumiya, M., Yamamoto, T., & Sadato, N. (2021).
1182 Neural substrates for sharing intention in action during face-to-face imitation. *NeuroImage*,
1183 233, 117916. <https://doi.org/10.1016/j.neuroimage.2021.117916>

1184 Molenberghs, P., Cunnington, R., & Mattingley, J. B. (2012). Brain regions with mirror properties:
1185 A meta-analysis of 125 human fMRI studies. *Neuroscience and Biobehavioral Reviews*,
1186 36(1), 341–349. <https://doi.org/10.1016/j.neubiorev.2011.07.004>

1187 Molenberghs, P., Johnson, H., Henry, J. D., & Mattingley, J. B. (2016). Understanding the minds of
1188 others: A neuroimaging meta-analysis. *Neuroscience & Biobehavioral Reviews*, 65, 276–
1189 291. <https://doi.org/10.1016/j.neubiorev.2016.03.020>

1190 Möttönen, R., Farmer, H., & Watkins, K. E. (2016). Neural basis of understanding communicative
1191 actions: Changes associated with knowing the actor’s intention and the meanings of the
1192 actions. *Neuropsychologia*, 81, 230–237.
1193 <https://doi.org/10.1016/j.neuropsychologia.2016.01.002>

1194 Nagels, A., Kircher, T., Steines, M., & Straube, B. (2015). Feeling addressed! The role of body
1195 orientation and co-speech gesture in social communication. *Human Brain Mapping*, 36(5),
1196 1925–1936. <https://doi.org/10.1002/hbm.22746>

1197 Narayanan, N. S., & Laubach, M. (2006). Top-down control of motor cortex ensembles by
1198 dorsomedial prefrontal cortex. *Neuron*, 52(5), 921–931.
1199 <https://doi.org/10.1016/j.neuron.2006.10.021>

1200 Newmannorlund, R., Bosga, J., Meulenbroek, R., & Bekkering, H. (2008). Anatomical substrates of
1201 cooperative joint-action in a continuous motor task: Virtual lifting and balancing.
1202 *NeuroImage*, 41(1), 169–177. <https://doi.org/10.1016/j.neuroimage.2008.02.026>

1203 Nili, H., Wingfield, C., Walther, A., Su, L., Marslen-Wilson, W., & Kriegeskorte, N. (2014). A
1204 toolbox for representational similarity analysis. *PLoS Computational Biology*, 10(4),
1205 e1003553. <https://doi.org/10.1371/journal.pcbi.1003553>

- 1206 Ninomiya, T., Noritake, A., Kobayashi, K., & Isoda, M. (2020). A causal role for frontal cortico-
 1207 cortical coordination in social action monitoring. *Nature Communications*, 11(1), 5233.
 1208 <https://doi.org/10.1038/s41467-020-19026-y>
- 1209 Ninomiya, T., Noritake, A., Ullsperger, M., & Isoda, M. (2018). Performance monitoring in the
 1210 medial frontal cortex and related neural networks: From monitoring self actions to
 1211 understanding others' actions. *Neuroscience Research*, 137, 1–10.
 1212 <https://doi.org/10.1016/j.neures.2018.04.004>
- 1213 Orban, G. A., Sepe, A., & Bonini, L. (2021). Parietal maps of visual signals for bodily action
 1214 planning. *Brain Structure and Function*, 226(9), 2967–2988. [https://doi.org/10.1007/s00429-](https://doi.org/10.1007/s00429-021-02378-6)
 1215 [021-02378-6](https://doi.org/10.1007/s00429-021-02378-6)
- 1216 Otti, A., Wohlschlaeger, A. M., & Noll-Hussong, M. (2015). Is the Medial Prefrontal Cortex
 1217 Necessary for Theory of Mind? *PLOS ONE*, 10(8), e0135912.
 1218 <https://doi.org/10.1371/journal.pone.0135912>
- 1219 Patel, G. H., Sestieri, C., & Corbetta, M. (2019). The evolution of the temporoparietal junction and
 1220 posterior superior temporal sulcus. *Cortex*, 118, 38–50.
 1221 <https://doi.org/10.1016/j.cortex.2019.01.026>
- 1222 Patri, J.-F., Cavallo, A., Pullar, K., Soriano, M., Valente, M., Koul, A., Avenanti, A., Panzeri, S., &
 1223 Becchio, C. (2020). Transient Disruption of the Inferior Parietal Lobule Impairs the Ability
 1224 to Attribute Intention to Action. *Current Biology: CB*, 30(23), 4594–4605.e7.
 1225 <https://doi.org/10.1016/j.cub.2020.08.104>
- 1226 Peelen, M. V., & Downing, P. E. (2005). Selectivity for the Human Body in the Fusiform Gyrus.
 1227 *Journal of Neurophysiology*, 93(1), 603–608. <https://doi.org/10.1152/jn.00513.2004>
- 1228 Peirce, J. W., Hirst, R. J., & MacAskill, M. R. (2022). *Building Experiments in PsychoPy. 2nd Edn.*
 1229 Sage.
- 1230 Perrett, D. I., Harries, M. H., Bevan, R., Thomas, S., Benson, P. J., Mistlin, A. J., Chitty, A. J.,
 1231 Hietanen, J. K., & Ortega, J. E. (1989). Frameworks of analysis for the neural representation
 1232 of animate objects and actions. *Journal of Experimental Biology*, 146(1), 87–113.
 1233 <https://doi.org/10.1242/jeb.146.1.87>
- 1234 Pitcher, D., & Ungerleider, L. G. (2021). Evidence for a Third Visual Pathway Specialized for
 1235 Social Perception. *Trends in Cognitive Sciences*, 25(2), 100–110.
 1236 <https://doi.org/10.1016/j.tics.2020.11.006>
- 1237 Popal, H., Wang, Y., & Olson, I. R. (2019). A Guide to Representational Similarity Analysis for
 1238 Social Neuroscience. *Social Cognitive and Affective Neuroscience*, 14(11), 1243–1253.
 1239 <https://doi.org/10.1093/scan/nsz099>
- 1240 Prinsen, J., & Alaerts, K. (2019). Eye contact enhances interpersonal motor resonance: Comparing
 1241 video stimuli to a live two-person action context. *Social Cognitive and Affective*
 1242 *Neuroscience*, 14(9), 967–976. <https://doi.org/10.1093/scan/nsz064>
- 1243 Pryke, A., Jayachandran, K., & Martin, A. K. (2025). The causal neural substrates underpinning
 1244 prospective and retrospective sense of agency. *Cortex*, 188, 53–67.
 1245 <https://doi.org/10.1016/j.cortex.2025.04.014>
- 1246 Quesque, F., & Brass, M. (2019). The Role of the Temporoparietal Junction in Self-Other
 1247 Distinction. *Brain Topography*, 32(6), 943–955. [https://doi.org/10.1007/s10548-019-00737-](https://doi.org/10.1007/s10548-019-00737-5)
 1248 [5](https://doi.org/10.1007/s10548-019-00737-5)
- 1249 Randriamihamison, N., Vialaneix, N., & Neuvial, P. (2021). Applicability and Interpretability of
 1250 Ward's Hierarchical Agglomerative Clustering With or Without Contiguity Constraints.
 1251 *Journal of Classification*, 38(2), 363–389. <https://doi.org/10.1007/s00357-020-09377-y>
- 1252 Rauchbauer, B., Majdandžić, J., Hummer, A., Windischberger, C., & Lamm, C. (2015). Distinct
 1253 neural processes are engaged in the modulation of mimicry by social group-membership and
 1254 emotional expressions. *Cortex*, 70, 49–67. <https://doi.org/10.1016/j.cortex.2015.03.007>
- 1255 Reader, A. T., Royce, B. P., Marsh, J. E., Chivers, K., & Holmes, N. P. (2018). Repetitive
 1256 transcranial magnetic stimulation reveals a role for the left inferior parietal lobule in

1257 matching observed kinematics during imitation. *European Journal of Neuroscience*, 47(8),
 1258 918–928. <https://doi.org/10.1111/ejn.13886>
 1259 Redcay, E., & Schilbach, L. (2019). Using second-person neuroscience to elucidate the mechanisms
 1260 of social interaction. *Nature Reviews Neuroscience*, 20(8), 495–505.
 1261 <https://doi.org/10.1038/s41583-019-0179-4>
 1262 Rizzolatti, G., & Sinigaglia, C. (2016). The mirror mechanism: A basic principle of brain function.
 1263 *Nature Reviews. Neuroscience*, 17(12), 757–765. <https://doi.org/10.1038/nrn.2016.135>
 1264 Sagliano, L., Vela, M., Trojano, L., & Conson, M. (2019). The role of the right premotor cortex and
 1265 temporo-parietal junction in defensive responses to visual threats. *Cortex*, 120, 532–538.
 1266 <https://doi.org/10.1016/j.cortex.2019.08.005>
 1267 Saito, D. N., Tanabe, H. C., Izuma, K., Hayashi, M. J., Morito, Y., Komeda, H., Uchiyama, H.,
 1268 Kosaka, H., Okazawa, H., Fujibayashi, Y., & Sadato, N. (2010). “Stay Tuned”: Inter-
 1269 Individual Neural Synchronization During Mutual Gaze and Joint Attention. *Frontiers in*
 1270 *Integrative Neuroscience*, 4. <https://doi.org/10.3389/fnint.2010.00127>
 1271 Saporta, N., Scheele, D., Lieberz, J., Nevat, M., Kanterman, A., Hurlmann, R., & Shamay-Tsoory,
 1272 S. G. (2022). Altered activation in the action observation system during synchronization in
 1273 high loneliness individuals. *Cerebral Cortex*, 33(2), 385–402.
 1274 <https://doi.org/10.1093/cercor/bhac073>
 1275 Saxe, R. (2006). Uniquely human social cognition. *Current Opinion in Neurobiology*, 16(2), 235–
 1276 239. <https://doi.org/10.1016/j.conb.2006.03.001>
 1277 Saxe, R., & Wexler, A. (2005). Making sense of another mind: The role of the right temporo-
 1278 parietal junction. *Neuropsychologia*, 43(10), 1391–1399.
 1279 <https://doi.org/10.1016/j.neuropsychologia.2005.02.013>
 1280 Schilbach, L., Wilms, M., Eickhoff, S. B., Romanzetti, S., Tepest, R., Bente, G., Shah, N. J., Fink,
 1281 G. R., & Vogeley, K. (2010). Minds Made for Sharing: Initiating Joint Attention Recruits
 1282 Reward-related Neurocircuitry. *Journal of Cognitive Neuroscience*, 22(12), 2702–2715.
 1283 <https://doi.org/10.1162/jocn.2009.21401>
 1284 Schmidt, S. N. L., Hass, J., Kirsch, P., & Mier, D. (2021). The human mirror neuron system-A
 1285 common neural basis for social cognition? *Psychophysiology*, 58(5), e13781.
 1286 <https://doi.org/10.1111/psyp.13781>
 1287 Schurz, M., Aichhorn, M., Martin, A., & Perner, J. (2013). Common brain areas engaged in false
 1288 belief reasoning and visual perspective taking: A meta-analysis of functional brain imaging
 1289 studies. *Frontiers in Human Neuroscience*, 7, 712.
 1290 <https://doi.org/10.3389/fnhum.2013.00712>
 1291 Schurz, M., Maliske, L., & Kanske, P. (2020). Cross-network interactions in social cognition: A
 1292 review of findings on task related brain activation and connectivity. *Cortex*, 130, 142–157.
 1293 <https://doi.org/10.1016/j.cortex.2020.05.006>
 1294 Schurz, M., Radua, J., Aichhorn, M., Richlan, F., & Perner, J. (2014). Fractionating theory of mind:
 1295 A meta-analysis of functional brain imaging studies. *Neuroscience and Biobehavioral*
 1296 *Reviews*, 42, 9–34. <https://doi.org/10.1016/j.neubiorev.2014.01.009>
 1297 Shepherd, S. V., & Freiwald, W. A. (2018). Functional Networks for Social Communication in the
 1298 Macaque Monkey. *Neuron*, 99(2), 413–420.e3. <https://doi.org/10.1016/j.neuron.2018.06.027>
 1299 Simos, P. G., Kavroulakis, E., Maris, T., Papadaki, E., Boursianis, T., Kalaitzakis, G., & Savaki, H.
 1300 E. (2017). Neural foundations of overt and covert actions. *NeuroImage*, 152, 482–496.
 1301 <https://doi.org/10.1016/j.neuroimage.2017.03.036>
 1302 Sladky, R., Friston, K. J., Tröstl, J., Cunningham, R., Moser, E., & Windischberger, C. (2011). Slice-
 1303 timing effects and their correction in functional MRI. *NeuroImage*, 58(2), 588–594.
 1304 <https://doi.org/10.1016/j.neuroimage.2011.06.078>
 1305 Sliwa, J., & Freiwald, W. A. (2017). A dedicated network for social interaction processing in the
 1306 primate brain. *Science (New York, N.Y.)*, 356(6339), Article 6339.
 1307 <https://doi.org/10.1126/science.aam6383>

- Sperduti, M., Guionnet, S., Fossati, P., & Nadel, J. (2014). Mirror Neuron System and Mentalizing System connect during online social interaction. *Cognitive Processing*, 15(3), 307–316. <https://doi.org/10.1007/s10339-014-0600-x>
- Spunt, R. P., Satpute, A. B., & Lieberman, M. D. (2011). Identifying the What, Why, and How of an Observed Action: An fMRI Study of Mentalizing and Mechanizing during Action Observation. *Journal of Cognitive Neuroscience*, 23(1), 63–74. <https://doi.org/10.1162/jocn.2010.21446>
- Sun, D., Shao, R., Wang, Z., & Lee, T. M. C. (2018). Perceived Gaze Direction Modulates Neural Processing of Prosocial Decision Making. *Frontiers in Human Neuroscience*, 12, 52. <https://doi.org/10.3389/fnhum.2018.00052>
- Szczepanski, S. M., Pinsk, M. A., Douglas, M. M., Kastner, S., & Saalmann, Y. B. (2013). Functional and structural architecture of the human dorsal frontoparietal attention network. *Proceedings of the National Academy of Sciences*, 110(39), 15806–15811. <https://doi.org/10.1073/pnas.1313903110>
- Thornton, M. A., Weaverdyck, M. E., & Tamir, D. I. (2019). The Social Brain Automatically Predicts Others' Future Mental States. *The Journal of Neuroscience*, 39(1), 140–148. <https://doi.org/10.1523/JNEUROSCI.1431-18.2018>
- Tognoli, E., Lagarde, J., DeGuzman, G. C., & Kelso, J. A. S. (2007). The phi complex as a neuromarker of human social coordination. *Proceedings of the National Academy of Sciences*, 104(19), 8190–8195. <https://doi.org/10.1073/pnas.0611453104>
- Urgen, B. A., & Orban, G. A. (2021). The unique role of parietal cortex in action observation: Functional organization for communicative and manipulative actions. *NeuroImage*, 237, 118220. <https://doi.org/10.1016/j.neuroimage.2021.118220>
- Urgen, B. A., Pehlivan, S., & Saygin, A. P. (2019). Distinct representations in occipito-temporal, parietal, and premotor cortex during action perception revealed by fMRI and computational modeling. *Neuropsychologia*, 127, 35–47. <https://doi.org/10.1016/j.neuropsychologia.2019.02.006>
- Wang, Y., Ramsey, R., & De C. Hamilton, A. F. (2011). The Control of Mimicry by Eye Contact Is Mediated by Medial Prefrontal Cortex. *The Journal of Neuroscience*, 31(33), 12001–12010. <https://doi.org/10.1523/JNEUROSCI.0845-11.2011>
- Wheatley, T., Milleville, S. C., & Martin, A. (2007). Understanding Animate Agents: Distinct Roles for the Social Network and Mirror System. *Psychological Science*, 18(6), 469–474. <https://doi.org/10.1111/j.1467-9280.2007.01923.x>
- Wittmann, M. K., Trudel, N., Trier, H. A., Klein-Flügge, M. C., Sel, A., Verhagen, L., & Rushworth, M. F. S. (2021). Causal manipulation of self-other mergence in the dorsomedial prefrontal cortex. *Neuron*, 109(14), 2353–2361.e11. <https://doi.org/10.1016/j.neuron.2021.05.027>
- Wong, A. L., Jax, S. A., Smith, L. L., Buxbaum, L. J., & Krakauer, J. W. (2019). Movement Imitation via an Abstract Trajectory Representation in Dorsal Premotor Cortex. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 39(17), 3320–3331. <https://doi.org/10.1523/JNEUROSCI.2597-18.2019>
- Woolrich, M. (2008). Robust group analysis using outlier inference. *NeuroImage*, 41(2), 286–301. <https://doi.org/10.1016/j.neuroimage.2008.02.042>
- Woolrich, M. W., Ripley, B. D., Brady, M., & Smith, S. M. (2001). Temporal Autocorrelation in Univariate Linear Modeling of FMRI Data. *NeuroImage*, 14(6), 1370–1386. <https://doi.org/10.1006/nimg.2001.0931>
- Woźniak, M., Schmidt, T. T., Wu, Y.-H., Blankenburg, F., & Hohwy, J. (2022). Differences in working memory coding of biological motion attributed to oneself and others. *Human Brain Mapping*, 43(12), 3721–3734. <https://doi.org/10.1002/hbm.25879>
- Yoshida, K., Saito, N., Iriki, A., & Isoda, M. (2012). Social error monitoring in macaque frontal cortex. *Nature Neuroscience*, 15(9), 1307–1312. <https://doi.org/10.1038/nn.3180>